

Quadratic hulls on boxes: valid inequalities and semidefinite representability

Abstract

Convexifying the products of a few bounded continuous variables is a basic step in relaxing a nonconvex quadratic program. We study which valid inequalities are missing from existing local relaxations and when the exact quadratic moment hull admits a finite semidefinite lift. An explicit rational three-variable example separates a disjoint-support affine-square certificate system, even when strengthened by the specified extended-triangle second-order-cone inequalities. The example belongs to a family of valid quadratic inequalities whose entire parameter range is enforced by one positive-semidefinite block of order five and six nonnegative auxiliary variables. We prove exposedness in the strict parameter region and extremality in several boundary regimes. In three variables, we also classify every extreme ray with positive square coefficients and positive values at all cube vertices: apart from affine squares, these rays are generated by strict family members up to cube symmetry. We establish a sharp representability boundary: the full quadratic moment hull of a box has a finite semidefinite lift in dimension at most three, but has none in dimension at least four. The obstruction extends to polytopes with a simple vertex and to sparse joint hulls whose positive-loop graph contains a K_4 minor; components of at most three positive-loop vertices admit explicit lifts. The nonrepresentability proofs transfer known results about completely positive maps over real closed fields to these hulls. Archived computations show that selective family blocks can match exact triple lifts on constructed instances, while offering little improvement on most tested benchmark instances.

Keywords: quadratic optimization; convex hull; semidefinite relaxation; valid inequalities; spectral shadow; sparse optimization.

1 Introduction

A quadratic objective on a box depends on the first and second moments of its variables. Replacing the possible moment vectors by their convex hull therefore gives an exact convex formulation. The difficulty is describing that hull. Even a relaxation built from many valid inequalities can omit a nonnegative quadratic with only three variables. Conversely, the existence of an exact formulation in three variables does not imply that a larger finite semidefinite formulation will be exact in four variables.

This paper studies these two issues through the cone of quadratics nonnegative on $[0, 1]^n$. We first identify a family of valid three-variable inequalities that recent semidefinite and second-order-cone relaxations miss. We then give an exact, small semidefinite formulation of that family. The same cone provides the setting for a dimension threshold: the full quadratic moment hull has a finite semidefinite lift precisely when $n \leq 3$. The negative result also holds when positive surplus is allowed on the diagonal moments. This variant is the appropriate joint hull for objectives whose square coefficients are nonnegative.

The connection between the two parts is concrete. A nonnegative quadratic is a valid linear inequality for the moment hull. A certificate system describes a subcone of these quadratics, and its

dual defines a relaxation. The new family fills a proved gap in one such system. The representability result determines where any finite semidefinite system, regardless of its chosen certificates or auxiliary variables, must cease to be exact.

1.1 Relation to prior work

Standard box relaxations combine the Shor moment-matrix condition [18] with McCormick inequalities for bilinear products [13]. The continuous polynomial form of the reformulation–linearization technique (RLT) provides a systematic framework for linearizing products of bound inequalities [17].

Anstreicher and Burer gave exact completely positive descriptions of quadratic moment hulls on low-dimensional polytopes [1]. Triangulating a polytope of dimension at most three produces completely positive blocks of order at most four. At these orders, complete positivity is equivalent to positive semidefiniteness and entrywise nonnegativity [8, 12]. Thus the exact three-variable box hull already has a finite semidefinite lift. We use that construction both as a mathematical reference and as an exact local strengthening in the archived computational comparisons. We do not propose a first exact formulation for this hull.

Burer and Dong developed recursive descriptions and separation methods for the related box completely positive and copositive cones [5]. An exact separation procedure and a finite semidefinite lift are different properties. Our impossibility theorem concerns the latter; it does not prevent separation or other exact optimization methods.

Recent work seeks formulations that exploit sparse objectives and small local blocks. Khajavirad’s disjoint-support construction uses sums of affine squares weighted by products of box literals, with the affine factor involving only variables absent from the weight [11]. In three variables with positive square coefficients, the resulting extended formulation consists of 27 localizing matrices. Anstreicher and Puges strengthen the usual three-variable inequalities using a trilinear auxiliary variable and second-order-cone inequalities [2]. We give a single rational moment functional feasible for both systems, including all coordinate complements and permutations of the latter inequalities, but violating a nonnegative quadratic. This is a comparison against the specified systems, rather than a claim about all possible SOS or SOC relaxations.

Sign structure supplies an additional point of comparison. Burer, Natarajan, and Willemsen prove exactness of a semidefinite relaxation for submodular box quadratics in at most three variables [6]. Combined with cube symmetries, this result settles the three-variable disjoint-support certificate question for quadratics with nonnegative square coefficients whenever the product of the three mixed coefficients is nonpositive. The new inequalities occupy the remaining sign class. This restriction helps explain why they require three variables and why two-variable restrictions do not detect them.

Hildebrand’s rank-one subtraction criterion for copositive matrices [10] supplies a further geometric tool. Applied through the cube vertices, it implies that the zeros of a non-square extreme quadratic must affinely span the box. We combine this criterion with an explicit analysis of convex facet restrictions and a classification of cube-edge contact graphs. Together with the submodular duality argument, these steps classify every extreme ray with positive square coefficients and no vertex zero. The subtraction criterion is prior work; the facet analysis and the resulting box classification are the developments proved here.

For the negative representability result, the essential external tool is the characterization of spectrahedral shadows by preservation under completely positive maps over real closed fields due to Boudier, Kummer, and Thom [3]. Their construction uses the Horn copositive matrix to exhibit an obstruction to semidefinite representability. We transfer that obstruction to quadratics on a box by using the constant monomial as one copositive coordinate and placing the other four coordinates near a box vertex. The transfer rules out every finite semidefinite lift, rather than only a doubly

nonnegative approximation. We also extend the argument to polytopes with a simple vertex and to sparse positive-loop graphs containing a K_4 minor. These conclusions use the existing preservation theorem; the application to the box and the stated sparse hulls is the result proved here.

Nishijima proves that the copositive and completely positive cones over any symmetric cone of rank at least five are not spectrahedral shadows [14, Theorem 3.2]. The present paper uses the explicit box embedding and graph reductions to reach the moment hulls studied here.

1.2 Contributions and scope

The paper makes the following contributions.

1. An exact rational certificate proves that the three-variable disjoint-support system misses a quadratic with strictly positive square coefficients. The same certificate satisfies the switched SOC strengthening of Anstreicher and Puges. A direct proof using five edge zeros explains the failure without a numerical optimization argument (Section 3).
2. A parameter family generalizes this example. Its validity follows from an explicit polynomial identity. In an open parameter region its members generate exposed extreme rays of the nonnegative-quadratic cone and admit no disjoint-support certificate. All inequalities in the larger validity domain are enforced by one positive semidefinite block of order five and six nonnegative scalars, using only first and second moments (Section 4).
3. We establish the closedness and dual interpretation of the disjoint-support cone, relate diagonal caps to the distinction between the full and positive-loop hulls, and prove the sign restriction and boundary extreme-ray results needed to state a precise three-variable completeness question. We further classify every extreme ray with positive square coefficients and positive values at all cube vertices, reducing the remaining completeness question to vertex-zero rays (Section 5). The completeness question remains separate from the validity and enforcement theorems.
4. We prove the sharp dimension threshold for finite semidefinite lifts of the full box moment hull, the positive-loop hull, and their dual cones. We prove a local geometric extension at simple vertices and show that every proper face of the four-variable box hull nevertheless has such a lift (Section 6).
5. For sparse moment hulls, we prove a K_4 -minor obstruction on the positive-loop subgraph and construct finite lifts when every positive-loop component has at most three vertices. The construction gives an explicit size bound in terms of a tree decomposition after the component boundaries are made cliques (Section 7). Archived computations compare the new inequalities with the named relaxations and exact local hulls, with separate conclusions for constructed and benchmark instances (Section 8).

To the best of our knowledge, the family and its order-five formulation, the exact separating certificate against the two named recent systems, the classification of three-variable extreme rays with positive square coefficients and no vertex zero, and the stated box and sparse-graph representability consequences have not appeared in the prior work reviewed here. The proofs below specify these claims independently of priority. In particular, neither the classical order-four cone identity nor the existing three-variable tetrahedral lift is a new result.

The practical use of the family is local convexification. Each selected triple contributes a block with no trilinear or higher-order moment variables, so the inequalities can strengthen a relaxation that already shares first and second moments. The archived study contains constructed instances

where this formulation attains the bound of exact triple hulls with fewer selected blocks. On the tested natural instances, the base relaxations were often already exact, and the new family produced little additional improvement. These findings support selective use of the family; they do not establish a general speed advantage or a branch-and-bound performance improvement.

Section 2 fixes the moment and coefficient conventions. Sections 3 to 5 develop the three-variable inequalities and their geometry. Sections 6 and 7 treat finite representability. Section 8 presents the archived computational evidence. The appendices contain the exact moment certificate, boundary calculations, stationary-facet and contact-graph proofs, real closed field details, and separation and comparison tools.

2 Quadratic moments and certificate systems

2.1 The full hull and the positive-loop hull

Let $C_n = [0, 1]^n$. We write a quadratic as

$$q(x) = q_0 + \sum_{i=1}^n a_i x_i + \sum_{i=1}^n b_i x_i^2 + \sum_{1 \leq i < j \leq n} c_{ij} x_i x_j. \quad (1)$$

Thus c_{ij} is the coefficient of the monomial $x_i x_j$, rather than an off-diagonal entry of a symmetric matrix representing q . Define

$$\mathcal{P}_n = \{q : q(x) \geq 0 \text{ for every } x \in C_n\}, \quad \mathcal{P}_n^+ = \{q \in \mathcal{P}_n : b_i \geq 0 \text{ for every } i\}.$$

The corresponding moment sets are

$$\mathcal{Q}_n = \text{conv}\{(x, xx^\top) : x \in C_n\}, \quad \mathcal{H}_n = \mathcal{Q}_n + \{(0, \text{diag}(s)) : s \in \mathbb{R}_+^n\}. \quad (2)$$

The set $\mathcal{Q}_n$ is compact. The set $\mathcal{H}_n$ is closed because it is the sum of a compact set and a closed cone. A point (m, Y) in $\mathcal{Q}_n$ records the first and second moments of a finite probability law on the box. In $\mathcal{H}_n$, each diagonal second moment may also contain nonnegative slack. We use the name *positive-loop hull* for this second set: a loop is the graph coordinate representing a square, and its sign specifies the permitted direction of diagonal slack.

For a quadratic, its evaluation at moments is

$$\mathcal{L}_{m,Y}(q) = q_0 + \sum_i a_i m_i + \sum_i b_i Y_{ii} + \sum_{i < j} c_{ij} Y_{ij}. \quad (3)$$

Every $q \in \mathcal{P}_n$ gives a valid inequality $\mathcal{L}_{m,Y}(q) \geq 0$ for $\mathcal{Q}_n$. It is valid for $\mathcal{H}_n$ exactly when $q \in \mathcal{P}_n^+$. Indeed, evaluating at $(x, xx^\top)$ gives $q(x)$, while adding diagonal slack changes the value by $\sum_i b_i s_i$. Conversely, finite-dimensional separation shows that these valid inequalities describe the two closed convex sets.

These are joint hulls of all recorded products. A lift of an epigraph for one fixed objective need not describe the full moment hull. This distinction will matter in the nonrepresentability results.

2.2 Normalization and duality

Taking cone duals requires retaining the constant coordinate. Let

$$\hat{\mathcal{Q}}_n = \text{cone}\{(1, x, xx^\top) : x \in C_n\}, \quad \hat{\mathcal{H}}_n = \hat{\mathcal{Q}}_n + \{(0, 0, \text{diag}(s)) : s \in \mathbb{R}_+^n\}.$$

Both cones are closed. In the first cone, every coordinate is bounded in absolute value by its mass coordinate, so a convergent sequence with mass tending to zero has limit zero. The same argument in the second cone leaves only the displayed diagonal slack when the mass tends to zero. For positive mass, closure follows by rescaling to the compact set $\mathcal{Q}_n$ and extracting a convergent slack vector.

Under the pairing obtained from (3) by replacing q_0 with tq_0 , their dual cones are $\mathcal{P}_n$ and $\mathcal{P}_n^+$, respectively. Their mass-one slices are $\mathcal{Q}_n$ and $\mathcal{H}_n$. The zero-mass slack in $\widehat{\mathcal{H}}_n$ cannot be discarded when using homogeneous duality.

We call a set a *spectrahedral shadow*, or equivalently a set with a *finite semidefinite lift*, if it can be written as

$$\{y : \exists z, A_0 + \sum_i y_i A_i + \sum_j z_j B_j \succeq 0\}, \quad (4)$$

possibly with additional affine equalities. Several PSD blocks are allowed, since they can be combined in one block-diagonal matrix. The word “finite” places no polynomial bound on the number of auxiliaries or the block orders. Consequently, failure of finite representability is stronger than failure of a particular small relaxation. It is not a computational hardness theorem.

2.3 The three-variable disjoint-support system

For the remainder of the local certificate analysis, $n = 3$. Let W be the ten-dimensional vector space of quadratics, including their constants. Let V be the twenty-dimensional span of the monomials x^α with $\alpha \in \{0, 1, 2\}^3$ and at most one exponent equal to two. For disjoint subsets $A, B \subseteq \{1, 2, 3\}$, put

$$w_{A,B}(x) = \prod_{i \in A} x_i \prod_{j \in B} (1 - x_j), \quad F_{A,B} = \{1, 2, 3\} \setminus (A \cup B).$$

Let $v_F(x)$ list the constant one and the coordinates indexed by F . Define the cone

$$\mathcal{K}_D = \left\{ \sum_{A \cap B = \emptyset} w_{A,B}(x) v_{F_{A,B}}(x)^\top G_{A,B} v_{F_{A,B}}(x) : G_{A,B} \succeq 0 \right\} \subseteq V, \quad \mathcal{D}_3 = \mathcal{K}_D \cap W. \quad (5)$$

Each summand is nonnegative on the cube. The matrix representation permits any finite sum of affine squares on the unused coordinates. The requirement that the final sum belongs to W imposes cancellation of all monomials of degree greater than two; it must not be replaced by a degree restriction on each summand. In particular, $\mathcal{D}_3 \subseteq \mathcal{P}_3^+$: the coefficient of x_i^2 comes only from a nonnegative diagonal entry of a block whose unused coordinates include i .

The dual local relaxation consists of restrictions to W of functionals $\ell \in V^*$ satisfying

$$\ell(1) = 1, \quad M_{A,B}(\ell) := \ell(w_{A,B} v_{F_{A,B}} v_{F_{A,B}}^\top) \succeq 0 \quad (A \cap B = \emptyset). \quad (6)$$

Here ℓ acts entrywise on a polynomial matrix. Write $\mathcal{R}_D$ for the resulting set of quadratic moment pairs (m, Y) , with $m_i = \ell(x_i)$ and $Y_{ij} = \ell(x_i x_j)$. There are 27 blocks: eight of order one, twelve of order two, six of order three, and one of order four. The higher moments in $V \setminus W$ are auxiliary coordinates. They need not come from one probability law. Every point of $\mathcal{H}_3$ has such an extension; conversely, equality of the relaxation with that hull is the substantive exactness question. Section 5 proves the closedness and extension statements needed to use this primal–dual description without a hidden closure assumption.

2.4 Diagonal caps and cube symmetries

The full box hull satisfies the three inequalities

$$Y_{ii} \leq m_i \quad (i = 1, 2, 3), \quad (7)$$

which are the moment forms of $x_i(1-x_i) \geq 0$. We call them *diagonal caps*. They have negative square coefficients and are not valid for the unbounded positive-loop hull. Thus a proposed description of $\mathcal{Q}_3$ and a proposed description of $\mathcal{H}_3$ are different statements, even when all other inequalities are the same.

Coordinate permutations and complements $x_i \mapsto 1 - x_i$ preserve the cube. They induce affine maps on normalized first and second moments and linear maps on homogeneous moments. Complementing coordinate i , for example, sends

$$m_i \mapsto 1 - m_i, \quad Y_{ii} \mapsto 1 - 2m_i + Y_{ii}, \quad Y_{ij} \mapsto m_j - Y_{ij} \quad (j \neq i),$$

with simultaneous substitution when more than one coordinate is complemented. These symmetries preserve $\mathcal{Q}_3$, $\mathcal{H}_3$, and the disjoint-support certificate and moment systems. They provide the different orientations of the inequality family constructed below.

3 A gap in the disjoint-support relaxation

The disjoint-support construction includes an affine-square certificate on every face of the cube. These certificates do not suffice for all nonnegative quadratics, even when every square coefficient is strictly positive. The following example gives both a geometric obstruction and an exact moment certificate of the gap.

Theorem 3.1. *The quadratic*

$$p(x, y, z) = x^2 + y^2 + 9z^2 + 6xy - 12xz - 12yz - x - y + 9z + \frac{1}{4} \quad (8)$$

belongs to $\mathcal{P}_3^+$ but not to $\mathcal{D}_3$. Its minimum on C_3 is zero, and its zero set consists of the five points

$$\begin{aligned} a &= (\tfrac{1}{2}, 0, 0), & b &= (0, \tfrac{1}{2}, 0), \\ c &= (1, 0, \tfrac{1}{6}), & d &= (0, 1, \tfrac{1}{6}), & e &= (1, 1, \tfrac{5}{6}). \end{aligned} \quad (9)$$

Proof. The polynomial identity

$$\begin{aligned} p &= (xy + x + y - 3z - \tfrac{1}{2})^2 + 6z(1-x)(1-y) \\ &\quad + 3xy(1-x) + 2xy(1-y) + x^2y(1-y) \end{aligned} \quad (10)$$

proves nonnegativity on the cube. Its square coefficients are 1, 1, 9. At a zero, the third and fourth terms imply either $xy = 0$ or $x = y = 1$. If $y = 0$, the second term implies $z = 0$ or $x = 1$; the square then gives a or c . The case $x = 0$ gives b or d . If $x = y = 1$, the square gives e . Each of these five points is indeed a zero.

Suppose that p has a disjoint-support certificate. Since $p(0, 0, 0) = 1/4$, some individual term $w_{A,B}\ell^2$ is strictly positive at the origin. Its weight must have $A = \emptyset$, and its affine factor has the form $\ell = \alpha + \beta x + \gamma y + \delta z$ with $\alpha \neq 0$. Every term in the certificate is nonnegative on the cube, so this term vanishes at each point in (9).

The weight is strictly positive at a and b , for every possible B . Thus $\beta = \gamma = -2\alpha$. The support restriction now excludes x and y from B , so $B \subseteq \{z\}$. The weight is consequently strictly positive at c and e . Vanishing at c gives $\delta = 6\alpha$, whereas at e the affine factor has value 2α . This contradicts its required vanishing. $\square$

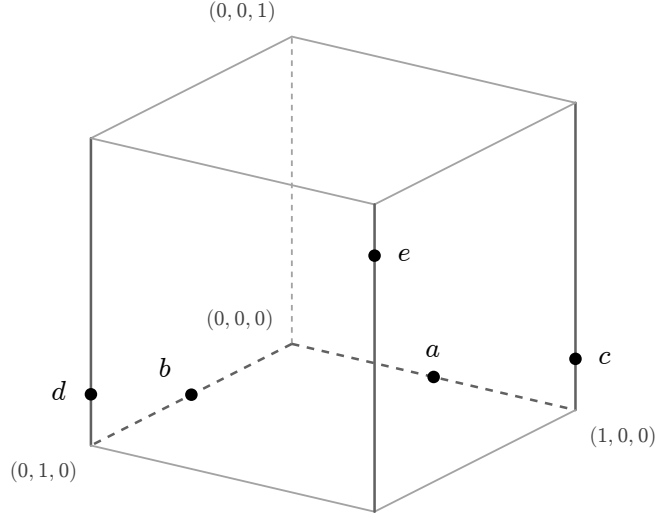


Figure 1: The five interior edge zeros of p in (9). Darker edges contain zeros; dashed lines indicate rear cube edges. These contacts explain the disjoint-support obstruction and determine the exposed ray in Theorem 4.2.

The identity (10) also identifies what the disjoint construction omits. It uses a bilinear expression inside a square, and a factor $y(1 - y)$ whose two literals concern the same coordinate. Neither is allowed in $\mathcal{K}_D$. The exclusion proof does not depend on a degree truncation: in three variables all permitted generators already have degree at most four, and all of them are included in $\mathcal{K}_D$.

The five zeros in Figure 1 explain the obstruction. An interior zero on an edge forces every certificate term whose weight is positive there to vanish through its affine factor. For a term positive at the origin, the first two zeros force both x and y to occur in that factor. The remaining zeros then impose inconsistent affine equations. This argument concerns the support restriction, rather than the size of a chosen semidefinite block.

Proposition 3.2 (Exact moment gap). *There is a linear functional Λ on V , with $\Lambda(1) = 1$, such that all 27 disjoint localizing matrices are positive definite and*

$$\Lambda(p) = -\frac{1}{40}. \quad (11)$$

Its quadratic moments also satisfy $Y_{ii} < m_i$ for all three coordinates. In particular, $p \notin \text{cl}(\mathcal{D}_3)$, and every feasible lower bound in the certificate problem

$$\sup\{\gamma : p - \gamma \in \mathcal{D}_3\}$$

is at most $-1/40$, although $\min_{C_3} p = 0$.

Proof. Use the rational moments in Table 4. Their localizing matrices and positive leading determinants are given in Proposition A.1. Since $\Lambda(w_{A,B}\ell^2)$ is the quadratic form of the corresponding positive definite matrix, Λ is nonnegative on $\mathcal{K}_D$. Substitution in (8) gives (11). Continuity therefore separates p from $\text{cl}(\mathcal{D}_3)$ as well as from $\mathcal{D}_3$. The diagonal upper-bound slacks are

$$m_x - Y_{xx} = m_y - Y_{yy} = \frac{649}{10000}, \quad m_z - Y_{zz} = \frac{696}{10000}.$$

If $p - \gamma \in \mathcal{D}_3$, applying Λ gives $-1/40 - \gamma \geq 0$. $\square$

This gap is not confined to quadratics with zeros. The polynomial $p + 1/80$ is at least $1/80$ throughout the cube, while $\Lambda(p + 1/80) = -1/80$. Thus even strict positivity does not make these disjoint certificates complete.

3.1 Comparison with two recent relaxations

For three positive loops, the matrices in the disjoint-support system are exactly the 27 matrices in equation (17) of Khajavirad [11]. Their orders are 4, 3, 2, 1, with respective multiplicities 1, 6, 12, 8. Consequently, Proposition 3.2 gives a negative answer to the three-variable exactness question for that system. This comparison uses the displayed matrices; it does not require a convention about the degree of the SOS certificate.

The same functional satisfies the second-order-cone strengthening of the extended triangle inequalities in Anstreicher and Puges [2]. Let $t = \Lambda(xyz)$. For distinct indices i, j, k , its two types of SOC inequalities are

$$t^2 \leq Y_{ii}Y_{jk}, \quad (Y_{ij} + t)^2 \leq Y_{ii}(Y_{jj} + 3Y_{jk}), \quad (12)$$

together with their coordinate permutations and complements. Their trilinear RLT inequalities are the eight nonnegative scalar localizing blocks. Proposition A.2 verifies all 24 instances of the first inequality and all 48 instances of the second, with strictly positive right-hand factors. The base moment matrix is positive definite, and the separate diagonal upper bounds hold by Proposition 3.2. The nonnegative scalar blocks imply the off-diagonal RLT and ordinary triangle inequalities; the SOC inequalities imply the three extended triangle families by Lemmas 4–5 of Anstreicher and Puges [2].

It follows that intersecting either named relaxation with $p \geq 0$ is a strict strengthening. The family introduced next gives a compact way to enforce this inequality and its parameter variations. This conclusion does not assert that one family block dominates either relaxation on its own. The established exact tetrahedral description of the three-variable hull already implies $p \geq 0$ [1].

4 A valid family and its compact semidefinite formulation

The example in Theorem 3.1 belongs to a parameter family with the same five-edge geometry. The validity domain is larger than the domain with five interior contacts. Retaining that larger domain is what makes exact enforcement by a small semidefinite formulation possible.

For $h \in \mathbb{R}$ and $d_1, d_2, d_3, k \geq 0$, set $L = h - d_1x - d_2y + d_3z$ and $D = d_1 + d_2 - h$, and define

$$q_{h,d,k} = L^2 + 2d_3kz(1 - x - y) + k(2D + k)xy. \quad (13)$$

In particular, h is unrestricted in sign. Let $\mathcal{F}$ denote the set of these polynomials, and let $\text{cone}(\mathcal{F})$ denote its conic hull. Scaling all five parameters by $\alpha \geq 0$ scales the polynomial by α^2 .

Theorem 4.1 (Validity). *Every member of $\mathcal{F}$ belongs to $\mathcal{P}_3^+$. Hence every inequality $\mathcal{L}_{m,Y}(q_{h,d,k}) \geq 0$ is valid on both $\mathcal{Q}_3$ and $\mathcal{H}_3$.*

Proof. The following polynomial identity holds for all real parameters:

$$\begin{aligned} q_{h,d,k} &= (L - kxy)^2 + 2d_3kz(1 - x)(1 - y) \\ &\quad + k(2d_1 + k)xy(1 - x) + 2kd_2xy(1 - y) + k^2x^2y(1 - y). \end{aligned} \quad (14)$$

To check it, expand the square and collect the remaining terms as $2kxyL + 2d_3kz(1 - x - y) + k(2D + k)xy - k^2x^2y^2$. After substituting L and D , this is precisely the sum of the four remaining terms in (14). Each term on the right is nonnegative on C_3 under the stated parameter restrictions. No sign condition on h is used. The square coefficients are d_i^2 , so at an atom with diagonal surplus $s_i \geq 0$ the evaluated inequality has value

$$q_{h,d,k}(x) + \sum_{i=1}^3 d_i^2 s_i \geq 0.$$

Convexity gives validity on $\mathcal{Q}_3$ and $\mathcal{H}_3$. □

Zero parameters are included in this result. For example, $k = 0$ gives an affine square, which already has a disjoint-support certificate. The following theorem concerns the strict parameter region where the family has five interior edge zeros.

Theorem 4.2 (Exposed rays missed by disjoint certificates). *Assume*

$$h, d_1, d_2, d_3, k > 0, \quad h < \min\{d_1, d_2\}, \quad D + k < d_3. \quad (15)$$

Then $q_{h,d,k}$ has exactly the five zeros

$$\begin{aligned} a &= (h/d_1, 0, 0), & b &= (0, h/d_2, 0), \\ c &= (1, 0, (d_1 - h)/d_3), & d &= (0, 1, (d_2 - h)/d_3), \\ e &= (1, 1, (D + k)/d_3). \end{aligned} \quad (16)$$

They lie in the relative interiors of their respective edges. The ray $\mathbb{R}_{+}q_{h,d,k}$ is exposed in $\mathcal{P}_3$, and $q_{h,d,k} \notin \mathcal{D}_3$.

Proof. The strict inequalities place each displayed zero in its edge interior. The third and fourth terms of (14) force any zero to satisfy $xy = 0$ or $x = y = 1$. If $y = 0$, the second term forces $z = 0$ or $x = 1$; the square then gives a or c . The case $x = 0$ gives b or d . When $x = y = 1$, the square gives e . Conversely all five points are zeros.

For exposedness, suppose that $r \in \mathcal{P}_3$ vanishes at these five points. Write its coefficients in the convention of Section 2.1:

$$r = r_0 + a_1x + a_2y + a_3z + b_1x^2 + b_2y^2 + b_3z^2 + c_{12}xy + c_{13}xz + c_{23}yz.$$

A univariate quadratic nonnegative on an interval and zero in its interior is a nonnegative multiple of the square centered at that zero; this includes the identically zero polynomial. Therefore

$$r(x, 0, 0) = b_1(x - h/d_1)^2, \quad r(0, y, 0) = b_2(y - h/d_2)^2,$$

with $b_1, b_2 \geq 0$. Comparing constants shows that for some $\lambda \geq 0$,

$$b_1 = \lambda d_1^2, \quad b_2 = \lambda d_2^2, \quad r_0 = \lambda h^2, \quad a_1 = -2\lambda h d_1, \quad a_2 = -2\lambda h d_2. \quad (17)$$

The vertical restriction through c is $b_3(z - (d_1 - h)/d_3)^2$. Evaluating at $z = 0$ and using $d_1 - h > 0$ gives $b_3 = \lambda d_3^2$. Vanishing of the three vertical derivatives at c, d, e gives

$$\begin{aligned} a_3 + c_{13} &= -2\lambda d_3(d_1 - h), \\ a_3 + c_{23} &= -2\lambda d_3(d_2 - h), \\ a_3 + c_{13} + c_{23} &= -2\lambda d_3(D + k). \end{aligned}$$

Solving yields

$$a_3 = 2\lambda d_3(h + k), \quad c_{13} = -2\lambda d_3(d_1 + k), \quad c_{23} = -2\lambda d_3(d_2 + k). \quad (18)$$

Finally, the restriction through e gives $r(1, 1, 0) = \lambda(D + k)^2$, and hence

$$c_{12} = \lambda(2d_1d_2 + k(2D + k)).$$

Every coefficient of r is now λ times that of $q_{h,d,k}$. The conclusion also holds when $\lambda = 0$, in which case all coefficients vanish. The functional

$$r \longmapsto r(a) + r(b) + r(c) + r(d) + r(e)$$

is nonnegative on $\mathcal{P}_3$ and has zero face exactly $\mathbb{R}_+q_{h,d,k}$. It therefore exposes this ray.

For disjoint-certificate exclusion, use a term positive at the origin, as in Theorem 3.1. Write its affine factor as $\alpha + \beta x + \gamma y + \delta z$, with $\alpha \neq 0$ and $A = \emptyset$. Its weight is positive at a and b , so $\beta = -\alpha d_1/h$ and $\gamma = -\alpha d_2/h$. These nonzero coefficients force $B \subseteq \{z\}$. The weight is then positive at c and e . The zero at c gives $\delta = \alpha d_3/h$, but the affine factor has value $\alpha k/h \neq 0$ at e , a contradiction. $\square$

The exposed-ray conclusion is stronger than validity. A member in (15) cannot be decomposed into nonproportional nonnegative quadratics. Its full mixed coefficients satisfy $c_{12} > 0$, $c_{13} < 0$, and $c_{23} < 0$. Their product is positive, which agrees with the sign restriction in Theorem 5.6. For $(h, d_1, d_2, d_3, k) = (1/2, 1, 1, 3, 1)$, this theorem gives the polynomial and zeros of Theorem 3.1.

4.1 Exact enforcement of all family inequalities

Let ℓ be a functional on quadratics, not necessarily a valid moment functional. Write $\ell(1) = t$, $\ell(x_i) = m_i$, and $\ell(x_i x_j) = Y_{ij}$, and form

$$b = \begin{pmatrix} -m_x \\ -m_y \\ m_z \\ -Y_{xy} \end{pmatrix}, \quad B = \begin{pmatrix} Y_{xx} & Y_{xy} & -Y_{xz} & Y_{xy} \\ Y_{xy} & Y_{yy} & -Y_{yz} & Y_{xy} \\ -Y_{xz} & -Y_{yz} & Y_{zz} & m_z - Y_{xz} - Y_{yz} \\ Y_{xy} & Y_{xy} & m_z - Y_{xz} - Y_{yz} & Y_{xy} \end{pmatrix}. \quad (19)$$

Both depend linearly on the retained first and second moments.

Theorem 4.3 (One order-five block and six auxiliaries). *For arbitrary t, m, Y , the inequalities*

$$\ell(q_{h,d,k}) \geq 0 \quad (h \in \mathbb{R}, d_1, d_2, d_3, k \geq 0)$$

hold if and only if there is a symmetric matrix $N \in \mathbb{S}^4$ satisfying

$$N_{ii} = 0, \quad N_{ij} \geq 0 \ (i < j), \quad \begin{pmatrix} t & b^T \\ b & B - N \end{pmatrix} \succeq 0. \quad (20)$$

In particular, at $t = 1$ they hold if and only if $B - bb^T \in \text{COP}_4$. The formulation uses one affine positive semidefinite block of order five, six scalar nonnegative variables, and no moments of degree greater than two.

Proof. Put $v = (d_1, d_2, d_3, k)^T$. Expanding (13) gives

$$\ell(q_{h,d,k}) = th^2 + 2hb^T v + v^T Bv. \quad (21)$$

At $v = 0$, nonnegativity for every h requires $t \geq 0$. If $t > 0$, minimization over the unrestricted scalar h gives

$$\min_{h \in \mathbb{R}} \ell(q_{h,d,k}) = v^T (B - t^{-1}bb^T)v, \quad h = -t^{-1}b^T v.$$

Thus the inequalities are equivalent to $B - t^{-1}bb^T \in \text{COP}_4$.

We use the classical order-four identity

$$\text{COP}_4 = \mathbb{S}_+^4 + \mathcal{N}_4, \quad \text{CP}_4 = \text{DNN}_4, \quad (22)$$

where $\mathcal{N}_4$ is the cone of symmetric entrywise nonnegative matrices [8, 12]. It gives an exact decomposition $B - t^{-1}bb^T = P + N$, with $P \succeq 0$ and $N \geq 0$ entrywise. Moving the diagonal part of N into P permits N to have zero diagonal. The Schur complement of $t > 0$ gives (20), and proves the converse as well.

If $t = 0$, the expression in (21) is nonnegative for every real h only if $b^T v = 0$ for all $v \geq 0$, or equivalently $b = 0$. The remaining condition is $B \in \text{COP}_4$. By (22), this again is exactly (20). Conversely, a positive semidefinite block with zero top-left entry has $b = 0$, so no additional case is needed. This also covers the zero functional and all parameter degeneracies. $\square$

The unrestricted sign of h is essential to this proof. Requiring $h \geq 0$ would generally prevent elimination at $h = -b^T v$ and would produce a different condition. The nonnegative matrix N is also essential: setting $N = 0$ can exclude valid cube moments. At the genuine cube atom $(x, y, z) = (1/2, 1/2, 0)$, the matrix $B - bb^T$ has a zero first diagonal entry but entry (1, 4) equal to $1/8$, so it is not positive semidefinite. All family inequalities are nevertheless valid there. The theorem is an SDP lift with auxiliary variables; it does not claim an unlifted LMI or minimality of the stated lift size.

4.2 Separation and extraction of a cut

At a normalized candidate moment point, let $A = B - bb^T$. The formulation in Theorem 4.3 can be tested as a feasibility SDP. Its equivalent separation problem is

$$\mu = \min\{\langle A, W \rangle : W \succeq 0, W \geq 0 \text{ entrywise}, \text{tr } W = 1\}. \quad (23)$$

Proposition 4.4. *Problem (23) has an attained optimum, and*

$$\mu = \min\{v^T A v : v \geq 0, \|v\|_2 = 1\}. \quad (24)$$

Thus $\mu < 0$ if and only if some family inequality is violated. A nonnegative vector v with $v^T A v < 0$, together with $h = -b^T v$, gives such an inequality.

Proof. The feasible set in (23) is nonempty and compact. For example, $(I + \mathbf{1}\mathbf{1}^T)/8$ is positive definite, strictly positive entrywise, and has trace one. By $\text{DNN}_4 = \text{CP}_4$, every feasible W has a finite decomposition $W = \sum_r u_r u_r^T$, with $u_r \geq 0$. Its trace condition gives $\sum_r \|u_r\|_2^2 = 1$. Discarding zero terms, its objective is the convex combination

$$\langle A, W \rangle = \sum_r \|u_r\|_2^2 \left(\frac{u_r}{\|u_r\|_2} \right)^T A \left(\frac{u_r}{\|u_r\|_2} \right).$$

Conversely each normalized nonnegative rank-one matrix is feasible. This proves (24); the last statement follows from (21) at $t = 1$. $\square$

A rank-one optimum exists, but an SDP solver need not return it. A negative objective matrix is an aggregate certificate. Extracting an individual cut from that matrix requires a completely positive factorization, after which at least one factor has negative quadratic value, or a separate search for a nonnegative vector. An exact alternative using the 15 nonempty coordinate supports is given in Proposition D.1. These statements concern exact arithmetic; a numerical implementation must certify its proposed violation.

For the rational point in Proposition 3.2, the parameters $h = 1/2$ and $v = (1, 1, 3, 1)^T$ give $-1/40$. Keeping this v fixed and optimizing h gives

$$h = \frac{2361}{5000}, \quad \ell(q_{h,d,k}) = -\frac{644321}{25000000}.$$

Hence the family block cuts off the point and strictly strengthens each relaxation compared in Section 3.1 when added to that relaxation.

4.3 Symmetry and selective use on larger problems

Coordinate permutations and complements preserve C_3 , and therefore preserve validity of the family. Their affine action on first and second moments is given in Section 2.4. The diagonal surplus at an atom is unchanged, so the same transformations preserve validity on $\mathcal{H}_3$ as well as on $\mathcal{Q}_3$. Swapping x and y merely exchanges d_1 and d_2 . Thus all symmetry copies can be imposed using at most 24 blocks: three choices of the distinguished coordinate and eight complement patterns.

For a sparse quadratic model, one may add a family block on a selected triple whose three edge moments are present. If an edge moment is absent, introducing it is a further modeling choice. One may instead generate individual inequalities by separation. The results above prove validity, exact enforcement of the selected family, and strict improvement for specified relaxation points. Completeness after every symmetry copy has been included is the question examined in the next section.

5 Structure of the three-variable cone

The family in Equation (13) supplies inequalities missing from the disjoint-support system. We now identify the part of the nonnegative quadratic cone in which such inequalities can be needed. The argument also separates two questions that a hull description must answer: whether the diagonal caps have been imposed, and whether all inequalities with nonnegative square coefficients have been found.

5.1 Closedness and exact extension duality

The higher-degree cancellations in (5) do not cause a closure problem. This fact will let us use zero sets to exclude membership in $\mathcal{D}_3$ without a separate limiting argument.

Lemma 5.1. *The cones $\mathcal{K}_D$ and $\mathcal{D}_3$ are closed. The quadratic*

$$q_*(x) = 8 + 4(x_1^2 + x_2^2 + x_3^2)$$

belongs to $W \cap \text{int } \mathcal{K}_D$.

Proof. For a weight $w = w_{A,B}$ and its free coordinates F , let

$$H_w = \int_{C_3} w(x) v_F(x) v_F(x)^\top dx.$$

This matrix is positive definite: the weight is positive in the interior of the cube, and a nonzero affine function cannot vanish there. If a sequence of polynomials in $\mathcal{K}_D$ converges in coefficients, choose a PSD Gram matrix $G_{j,w}$ for each weight in each polynomial. Their integrals are

$$\sum_w \text{tr}(H_w G_{j,w}).$$

These sums are bounded, and every summand is nonnegative. Since $\text{tr}(H_w G_{j,w}) \geq \lambda_{\min}(H_w) \text{tr}(G_{j,w})$, each sequence of Gram matrices is bounded. A subsequence converges simultaneously in all 27 blocks to PSD matrices representing the limiting polynomial. Thus $\mathcal{K}_D$ is closed, and so is its intersection with W .

Taking $G_w = I$ in every block gives

$$\sum_{A \cap B = \emptyset} w_{A,B} \left(1 + \sum_{i \in F_{A,B}} x_i^2 \right) = 8 + 4 \sum_i x_i^2.$$

The linear map from the Gram blocks to V is onto. Indeed, each monomial in V is a generator: use a constant square for a square-free monomial, and use x_i^2 as the square when its exponent is two. The image of the interior point consisting of identity Gram matrices is therefore an interior point of $\mathcal{K}_D$. $\square$

Proposition 5.2. *Every linear functional on W that is nonnegative on $\mathcal{D}_3$ has an extension to V that is nonnegative on $\mathcal{K}_D$. The moment relaxation $\mathcal{R}_D$ is closed, and, for a quadratic q ,*

$$\mathcal{L}_{m,Y}(q) \geq 0 \quad \text{for every } (m,Y) \in \mathcal{R}_D \quad \Longleftrightarrow \quad q \in \mathcal{D}_3. \quad (25)$$

Proof. Let $T : V^* \rightarrow W^*$ be restriction. Because $q_* \in \text{int } \mathcal{K}_D$, there is an $\epsilon > 0$ such that $q_* + \epsilon u \in \mathcal{K}_D$ whenever $\|u\| \leq 1$. Hence every $\ell \in \mathcal{K}_D^*$ satisfies $\epsilon \|\ell\| \leq \ell(q_*)$. A convergent sequence $T\ell_j$ with $\ell_j \in \mathcal{K}_D^*$ has bounded $\ell_j(q_*)$, and therefore a convergent subsequence of extensions. Consequently $T(\mathcal{K}_D^*)$ is closed. Its dual is $\mathcal{K}_D \cap W = \mathcal{D}_3$, so the bipolar theorem gives $T(\mathcal{K}_D^*) = \mathcal{D}_3^*$. This proves the extension statement and the closedness of the mass-one slice $\mathcal{R}_D$.

To account for normalization in (25), note that $\ell(1) \geq 0$ for $\ell \in \mathcal{K}_D^*$. Positive-mass functionals can be normalized. If $\ell(1) = 0$, add t times evaluation at any cube point, normalize, and then let $t \downarrow 0$. Nonnegativity on the mass-one slice therefore implies nonnegativity on the entire cone $\mathcal{K}_D^*$, which is equivalent to membership in $\mathcal{K}_D \cap W$. $\square$

The same proof applies to the disjoint-support construction in two variables. We write $\mathcal{D}_2$ and $\mathcal{R}_{D,2}$ for its quadratic cone and moment relaxation.

5.2 Diagonal caps and coordinate rounding

For a coordinate i , define the linear map

$$\rho_i f(x) = (1 - x_i) f(x)|_{x_i=0} + x_i f(x)|_{x_i=1}. \quad (26)$$

For a probability law, composing its moment functional with ρ_i replaces coordinate i , conditionally on the original point, by a Bernoulli variable with the same mean. It keeps first moments and mixed second moments, and sets the diagonal second moment to the first moment.

Proposition 5.3. *For every n ,*

$$\mathcal{P}_n = \mathcal{P}_n^+ + \text{cone}\{x_i(1 - x_i) : 1 \leq i \leq n\}, \quad (27)$$

$$\mathcal{Q}_n = \mathcal{H}_n \cap \{(m, Y) : Y_{ii} \leq m_i \text{ for every } i\}. \quad (28)$$

In (27), the positive-square-coefficient component can be chosen to have exactly the same mixed coefficients as the original quadratic.

Proof. If the coefficient b_i of x_i^2 is negative, then

$$\rho_i q = q + b_i x_i(1 - x_i).$$

The right side is nonnegative on the cube because it interpolates the two nonnegative restrictions of q . Its x_i^2 coefficient is zero, and no other square or mixed coefficient changes. Applying this operation to every negative b_i gives

$$q = q^+ + \sum_{b_i < 0} (-b_i) x_i(1 - x_i), \quad q^+ \in \mathcal{P}_n^+.$$

The reverse inclusion is immediate.

For the hull statement, write a point of the right side as a true moment point plus nonnegative diagonal slack. Take a finite probability law for the true moment point. Rounding coordinate i changes only its diagonal moment, from its original value to m_i . A mixture of the original and rounded laws realizes any value in this interval. Thus the assumed cap allows us to match the prescribed Y_{ii} . Treating the coordinates in turn changes no previously matched moment. This proves membership in $\mathcal{Q}_n$. The other inclusion follows from $x_i^2 \leq x_i$ on the cube. $\square$

The disjoint-support relaxation has the same rounding property even when its higher moments do not come from a probability law.

Lemma 5.4. *The map ρ_i preserves V and maps $\mathcal{K}_D$ into $\mathcal{K}_D$. If $(m, Y) \in \mathcal{R}_D$, then both the point obtained by replacing Y_{ii} by m_i and the point $(m, Y + tE_{ii})$, for $t \geq 0$, belong to $\mathcal{R}_D$. Consequently,*

$$\mathcal{R}_D = (\mathcal{R}_D \cap \{Y_{ii} \leq m_i \text{ for every } i\}) + \{(0, \text{diag}(s)) : s \geq 0\}. \quad (29)$$

Proof. On monomials in V , ρ_i replaces x_i^2 by x_i and fixes the others. For a generator $w_{A,B}L^2$, it fixes the generator if $i \in A \cup B$. If i is free, it gives the two generators

$$w_{A,B \cup \{i\}}(L|_{x_i=0})^2 + w_{A \cup \{i\}, B}(L|_{x_i=1})^2.$$

Thus $\ell \circ \rho_i$ is feasible whenever ℓ is feasible, and its quadratic moments have the claimed form.

Let δ_i extract the coefficient of the pure monomial x_i^2 . On a generator, this coefficient is zero if i is not free; otherwise it is $w_{A,B}(0)c_i^2 \geq 0$, where c_i is the coefficient of x_i in L . Therefore $\ell + t\delta_i$ remains feasible and adds only t to Y_{ii} . Rounding every coordinate for which $Y_{ii} > m_i$, then adding the removed diagonal slack, proves (29). $\square$

We use the two-variable exactness theorem of Anstreicher and Burer [1, Theorem 6]: the Shor moment matrix, the box RLT inequalities, and the diagonal caps describe $\mathcal{Q}_2$. Here the box RLT inequalities are the moment forms of products of two bound factors, such as $x_i(1 - x_j) \geq 0$.

Corollary 5.5. *The following statements hold.*

$$(i) \ \mathcal{D}_2 = \mathcal{P}_2^+.$$

(ii) If $q \in \mathcal{P}_3^+$ has a zero square coefficient, then $q \in \mathcal{D}_3$.

(iii) If $q \in \mathcal{P}_3$ is convex, then $q \in \mathcal{D}_3$.

(iv) For every $(m, Y) \in \mathcal{R}_D$, replacing any one Y_{ii} by m_i gives a point of $\mathcal{H}_3$. In particular, a point of $\mathcal{R}_D$ outside $\mathcal{H}_3$ satisfies all three caps strictly.

Proof. Every point of $\mathcal{R}_{D,2}$ satisfying the caps obeys the Shor and box RLT conditions, and hence lies in $\mathcal{Q}_2$. The two-variable version of Lemma 5.4 shows that any other feasible point is such a point plus nonnegative diagonal slack. Every member of $\mathcal{P}_2^+$ is therefore nonnegative on $\mathcal{R}_{D,2}$; exact extension duality gives (i).

If, for example, $b_3 = 0$, then $q = (1 - x_3)q|_{x_3=0} + x_3q|_{x_3=1}$. Both restrictions belong to $\mathcal{D}_2$ by (i). Multiplying their disjoint certificates by $1 - x_3$ and x_3 produces generators of $\mathcal{K}_D$, and the resulting sum is the quadratic q . This proves (ii).

For (iii), let x^* minimize q over the cube. Its first-order optimality conditions express

$$\nabla q(x^*)^\top (x - x^*) = \sum_{x_i^*=0} \lambda_i x_i + \sum_{x_i^*=1} \mu_i (1 - x_i), \quad \lambda_i, \mu_i \geq 0.$$

The Taylor expansion of q is the sum of this expression, the nonnegative constant $q(x^*)$, and a PSD quadratic form centered at x^* . Each term has a disjoint-support certificate.

For (iv), take any $q \in \mathcal{P}_3^+$. Its two endpoint restrictions in coordinate i belong to $\mathcal{D}_2$, so $\rho_i q \in \mathcal{K}_D$. A feasible extension ℓ therefore satisfies $(\ell \circ \rho_i)(q) \geq 0$. The rounded moments are nonnegative on all of $\mathcal{P}_3^+$, and hence belong to $\mathcal{H}_3$. If the original point has $Y_{ii} \geq m_i$, it is this rounded point plus nonnegative diagonal slack, and also belongs to $\mathcal{H}_3$. $\square$

5.3 Exactness for the nonpositive-product sign class

The mixed coefficients carry a simple invariant under complementation: the product $c_{12}c_{13}c_{23}$. A nonpositive product means that a cube symmetry makes the quadratic submodular, that is, makes all its mixed coefficients nonpositive.

Theorem 5.6. *If $q \in \mathcal{P}_3$ and $c_{12}c_{13}c_{23} \leq 0$, then*

$$q \in \mathcal{D}_3 + \text{cone}\{x_i(1 - x_i) : i = 1, 2, 3\}.$$

If also $b_i \geq 0$ for every i , then $q \in \mathcal{D}_3$.

Proof. If one mixed coefficient is zero, independently complement the endpoints of the other two edges, as necessary, to make those coefficients nonpositive. If all are nonzero and their product is negative, first make c_{12} and c_{13} negative by complementing coordinates 2 and 3 when needed. The invariant product then makes c_{23} negative as well. The disjoint cone and the square coefficients are preserved by these symmetries. We may therefore assume $c_{ij} \leq 0$ for all $i < j$.

The exactness theorem of Burer et al. [6, Theorem 1] states that, in at most three variables, a box quadratic with nonpositive off-diagonal matrix entries has the same minimum as its relaxation

$$\begin{pmatrix} 1 & m^\top \\ m & Y \end{pmatrix} \succeq 0, \quad Y \leq m\mathbf{1}^\top, \quad (30)$$

where the latter inequality is entrywise. The symmetric matrix in the objective has diagonal b_i and off-diagonal entry $c_{ij}/2$.

Suppose first that $b_i \geq 0$. Given $(m, Y) \in \mathcal{R}_D$, use (29) to write it as (m, Y') satisfying the caps plus nonnegative diagonal slack. The point (m, Y') is feasible for (30): its Shor matrix is the unweighted localizing block, and $Y'_{ij} \leq m_i$ for $i \neq j$ follows from the generator $x_i(1 - x_j)$. Exactness gives $\mathcal{L}_{m, Y'}(q) \geq \min_{C_3} q \geq 0$. Nonnegative square coefficients ensure that adding the diagonal slack cannot decrease this value. By Proposition 5.2, $q \in \mathcal{D}_3$.

For general square coefficients, use Proposition 5.3. Its explicit decomposition leaves the mixed coefficients unchanged, so the positive-square-coefficient component belongs to $\mathcal{D}_3$ by the preceding argument. $\square$

This theorem concerns the objective of a three-variable box problem. In a larger constrained model, an inequality on a triple can arise from a dual combination of the objective and constraints. The mixed coefficients of the objective alone then do not justify omitting a family block.

5.4 Extreme rays and the six relevant orientations

Let $\mathcal{F}$ be the family in Equation (13). For each of its 24 distinct cube-symmetry copies, define the closed cone $\mathcal{F}_g = \text{cl cone}\{q \circ g : q \in \mathcal{F}\}$. Set

$$\mathcal{S} = \mathcal{D}_3 + \sum_g \mathcal{F}_g, \quad \mathcal{R} = \mathcal{R}_D \cap \{(m, Y) : \mathcal{L}_{m, Y}(f) \geq 0 \text{ for all } f \in \mathcal{F}_g, g\}. \quad (31)$$

By Theorem 4.3, the additional inequalities defining $\mathcal{R}$ have a finite SDP representation.

Lemma 5.7. *The cone $\mathcal{S}$ is closed, and $\mathcal{S}$ is precisely the cone of quadratics nonnegative on $\mathcal{R}$. An extreme ray of $\mathcal{P}_3^+$ belongs to $\mathcal{S}$ if and only if it belongs to $\mathcal{D}_3$ or is a limit of normalized members of one family copy.*

Proof. The functional $J(q) = \int_{C_3} q(x) dx$ is strictly positive on every nonzero element of $\mathcal{P}_3$. The base $\{q \in \mathcal{P}_3 : J(q) = 1\}$ is compact: otherwise a sequence of unbounded norms, normalized in coefficient norm, would have a nonzero nonnegative limit with integral zero. Each summand in $\mathcal{S}$ is a closed subcone of $\mathcal{P}_3^+$. For a convergent sequence of sums, its total integral bounds the integral, and thus the norm, of every summand. Taking subsequences proves closedness of $\mathcal{S}$.

In homogeneous coordinates the moment cone defining $\mathcal{R}$ is the intersection of $\mathcal{D}_3^*$ with the $\mathcal{F}_g^*$. Its dual is the closure of the sum of their duals, which is $\mathcal{S}$ by closedness. The normalization argument in Proposition 5.2 applies here too: evaluation at a cube point is feasible for all the cones. This proves the asserted duality.

If an extreme ray of $\mathcal{P}_3^+$ is a sum of elements from its subcones, every nonzero summand lies on that ray. The unit-integral base of $\mathcal{F}_g$ is the convex hull of the compact set of limits of unit-integral members of its family copy. An extreme point of the larger base of $\mathcal{P}_3^+$ cannot be a nontrivial convex combination of these limits. This proves the characterization. $\square$

Corollary 5.8. *Every $q \in \mathcal{P}_3^+ \setminus \mathcal{D}_3$ has strictly positive square coefficients, an indefinite Hessian, and $c_{12}c_{13}c_{23} > 0$. After complementing at most one coordinate, its three mixed coefficients are strictly positive. Among the 24 family copies, only six can approach an extreme ray with this strict sign pattern: for each choice of the distinguished coordinate z , complement either z or both of the other coordinates.*

Proof. The first claims follow from Corollary 5.5 and Theorem 5.6. A positive product means that either all coefficients are positive or exactly two are negative. In the latter case, complement their common coordinate.

For the original family,

$$c_{xz} = -2d_3(d_1 + k) \leq 0, \quad c_{yz} = -2d_3(d_2 + k) \leq 0.$$

To obtain three strictly positive coefficients after complementation, the original c_{xy} must be positive. The pattern $(+, -, -)$ becomes $(+, +, +)$ exactly when the complemented set is $\{z\}$ or $\{x, y\}$. There are three choices of z . A sequence approaching a strictly positive pattern eventually has that same strict pattern, so no other orientation can contribute such a limit. $\square$

An extreme ray of $\mathcal{P}_3^+$ with all square coefficients strictly positive is also extreme in $\mathcal{P}_3$. To see this, let $q = p_1 + p_2$ with $p_j \in \mathcal{P}_3$. For sufficiently small $\epsilon > 0$, both $q + \epsilon(p_1 - p_2)$ and $q - \epsilon(p_1 - p_2)$ remain in $\mathcal{P}_3^+$: they are positive linear combinations of p_1, p_2 , and their square coefficients remain positive. Extremality in the smaller cone forces $p_1 - p_2$ to be proportional to q , and hence both summands lie on its ray.

The reduction is a restriction on possible missing inequalities; it does not prove that the six orientations suffice. The complete statement still to be decided is

$$\mathcal{P}_3^+ = \mathcal{S}. \quad (32)$$

It is equivalent to $\mathcal{H}_3 = \mathcal{R}$, and also to

$$\mathcal{Q}_3 = \mathcal{R} \cap \{Y_{ii} \leq m_i \text{ for every } i\}. \quad (33)$$

Indeed, one implication follows from Proposition 5.3. Conversely, if (33) holds, a point of $\mathcal{R}$ either has some $Y_{ii} \geq m_i$, and is in $\mathcal{H}_3$ by Corollary 5.5, or satisfies every cap and is in $\mathcal{Q}_3$. Thus $\mathcal{R} = \mathcal{H}_3$. Coordinate rounding maps $\mathcal{R}$ into $\mathcal{H}_3 \subseteq \mathcal{R}$ as well.

To prove (32), it is enough to treat extreme rays with all mixed coefficients strictly positive. The compact-base argument in Lemma 5.7, together with the finite-dimensional Krein–Milman theorem, reduces cone equality to its extreme rays, and Corollary 5.8 removes all other cases. The computational evidence in Section 8 supports this equality but does not establish it.

5.5 Zero-set obstructions and boundary extreme rays

Zero sets explain why weighted affine squares can fail. For a vertex v and a set of free coordinates F , put

$$U_F(v) = \{x \in C_3 : x_j \neq 1 - v_j \text{ for every } j \notin F\},$$

and let π_F be projection onto F .

Lemma 5.9. *Let $q \in \mathcal{P}_3$, with zero set Z , and suppose $q(v) > 0$ at a vertex v . If*

$$\pi_F(v) \in \text{aff}(\pi_F(Z \cap U_F(v))) \quad \text{for every } F \subseteq \{1, 2, 3\},$$

then $q \notin \mathcal{D}_3$. For $F = \emptyset$, the condition means that $Z \cap U_\emptyset(v)$ is nonempty.

Proof. In a disjoint certificate for q , every nonnegative summand vanishes on Z . Since $q(v) > 0$, some summand $w_{A,B}L^2$ has positive weight at v and $L(v) \neq 0$. The positive locus of this weight is $U_F(v)$ for its free coordinates F . Hence L vanishes on the projected zero set in the display, and on its affine hull. The hypothesis forces $L(v) = 0$, a contradiction. $\square$

Because $\mathcal{D}_3$ is closed, this criterion excludes limiting disjoint certificates as well as exact ones. By contrast, three interior edge zeros incident to one vertex cannot yield a missing ray.

Lemma 5.10. Suppose $q \in \mathcal{P}_3$, $q(0) > 0$, and $q(\alpha_i e_i) = 0$ with $0 < \alpha_i < 1$ for $i = 1, 2, 3$. Then

$$q(x) = q(0) \left[\left(1 - \sum_i \frac{x_i}{\alpha_i} \right)^2 + 2 \sum_{i < j} r_{ij} \frac{x_i x_j}{\alpha_i \alpha_j} \right], \quad r_{ij} \geq 0.$$

In particular, $q \in \mathcal{D}_3$; if it is extreme in $\mathcal{P}_3$, it is an affine square.

Proof. The nonnegative restriction to edge i has an interior double zero, and equals $q(0)(1 - x_i/\alpha_i)^2$. This fixes every coefficient except the three mixed coefficients and gives the stated representation with real r_{ij} . Evaluation at $(\alpha_i e_i + \alpha_j e_j)/2$ gives $q(0)r_{ij}/2 \geq 0$. The displayed terms have disjoint certificates. If some $r_{ij} > 0$, the decomposition contains the nonproportional nonnegative polynomials $x_i x_j$ and the affine square, and the ray is not extreme. $\square$

The exposed family rays of Theorem 4.2 persist at several parameter boundaries. Some edge zeros reach a vertex; their tangency conditions then matter as much as their values.

Theorem 5.11. Let $k > 0$ and $d_1, d_2 \geq 0$. A family member $q_{h,d,k}$ spans an extreme ray of $\mathcal{P}_3$, and hence of $\mathcal{P}_3^+$, in each of the following regimes:

- (i) $0 < h < \min(d_1, d_2)$ and $d_3 = d_1 + d_2 - h + k$;
- (ii) $h = 0$ and $d_3 \geq d_1 + d_2 + k$;
- (iii) $h = d_1 < d_2$ and $d_3 \geq d_2 + k$, or the same regime with d_1, d_2 exchanged.

The assertions include zero parameters d_1 or d_2 whenever their regime allows them.

Proof. The family is nonnegative by Theorem 4.1. Its edge restrictions and the vertex tangencies described in Appendix B give double zeros, including those at edge endpoints. In any decomposition into nonnegative quadratics, both summands inherit every zero and every listed tangent edge derivative. At an interior edge zero this follows from ordinary first-order optimality. At a vertex, their one-sided derivatives into the edge are nonnegative and sum to zero, so both vanish. The coefficient reconstruction in Appendix B shows that these linear contact conditions have a one-dimensional kernel in each regime. Thus both summands are multiples of $q_{h,d,k}$. $\square$

In regime (ii), a zero parameter produces a whole zero edge, for example $q(x, 0, 0) = d_1^2 x^2$ when $h = 0$. The proof does not require isolated zeros. Such members have a zero square coefficient and belong to $\mathcal{D}_3$ by Corollary 5.5. Extremality therefore does not by itself imply that a family member strengthens the disjoint-support system.

5.6 Extreme rays without vertex zeros

The strict five-contact family can be characterized without enumerating contact positions. Apart from affine squares, it accounts for every extreme ray with positive square coefficients and no vertex zero.

Write $q(x) = q_0 + a^\top x + x^\top Q x$, so that $Q_{ii} = b_i$ and $2Q_{ij} = c_{ij}$ in the coefficient convention of Section 2.

Theorem 5.12. Suppose that q generates an extreme ray of $\mathcal{P}_3$ and satisfies

$$b_i > 0, \quad q(v) > 0 \quad (v \in \{0, 1\}^3). \quad (34)$$

Then q is an affine square, or a cube symmetry transforms it into $q_{h,d,k}$ with

$$d_1, d_2, d_3, k > 0, \quad 0 < h < \min(d_1, d_2), \quad d_1 + d_2 - h + k < d_3.$$

Conversely, every symmetry copy of a family member in this parameter range generates an extreme ray and satisfies (34). An affine square satisfying (34) generates an extreme ray if and only if its affine factor takes both signs on the cube. In the non-square case, every 2×2 principal determinant of Q is strictly negative.

Proof outline. For a nonpositive mixed-coefficient product, the attained dual of the exact submodular SDP decomposes q into affine squares, nonnegative coordinate products, diagonal caps, and a nonnegative constant. Its extremality and positive vertex values force it to be an affine square. In the remaining sign class, complementing a coordinate if necessary makes all mixed coefficients positive. A non-square extreme ray cannot have a stationary zero of a positive-semidefinite facet restriction: all its zeros would lie in an affine plane, allowing subtraction of a nonzero affine square. Lemma F.2 proves this fact, including stationary zeros on facet boundaries. It excludes facet-interior zeros and forces positive inward derivatives at every edge zero. An extreme ray must therefore have at least five edge zeros: fewer give at most eight linear value and tangency equations in the ten quadratic coefficients, leaving a nontrivial two-sided perturbation.

The edges containing zeros form a subgraph of the cube. Two such edges meeting at a vertex must point toward the same Hamming-weight level; three meeting edges would decompose the quadratic into an affine square and nonnegative products. The remaining graph lies in the two outer three-edge stars and the central six-edge cycle. A direct classification of graphs with at least five edges leaves the five-contact family. Five central-cycle edges instead give an affine square plus a nonnegative multiple of

$$(1 - x_1)(1 - x_2)(1 - x_3) + x_1x_2x_3,$$

which is in $\mathcal{D}_3$; every other graph is excluded by two-variable nonnegativity. The complete perturbation and graph arguments are given in Appendix E. The converse follows from the five-contact extremality proof and its coefficient formulas. The same appendix proves the stated characterization of the affine-square rays. $\square$

Thus an extreme ray with positive square coefficients and no vertex zero is already accounted for by the disjoint cone or the family. Under these hypotheses, if one 2×2 principal determinant of Q is nonnegative, the ray must be an affine square.

By Corollary 5.8, an extreme ray missing from $\mathcal{S}$ has positive square coefficients. The extremality argument preceding (32) makes it extreme in $\mathcal{P}_3$ as well. The theorem therefore confines the full completeness question to rays with a vertex zero.

6 The dimension threshold for finite semidefinite lifts

The three-variable hull has a finite semidefinite lift, whether or not the inequalities studied above give a complete description. We now prove that this changes at four variables. The obstruction concerns every finite lift, with arbitrary auxiliary variables and matrix sizes.

Recall the definition of a spectrahedral shadow in (4). Linear images, linear preimages, affine sections, products, and Minkowski sums preserve this property. A closed convex cone is a spectrahedral shadow if and only if its dual is one [3, Remark 3.17].

6.1 Normalization and the positive diagonal directions

The constant coefficient of a quadratic must be retained when taking a dual. For a compact convex set K and a polyhedral cone D , write

$$\widehat{K + D} = \{(t, tk + d) : t \geq 0, k \in K, d \in D\}. \quad (35)$$

This is the closed homogenized cone; its height-zero section is $\{0\} \times D$.

Lemma 6.1. *Let K be nonempty, compact, and convex, and let D be a polyhedral cone. Then $K + D$ has a finite semidefinite lift if and only if $\widehat{K + D}$ has one.*

Proof. The cone in (35) is closed: in a convergent sequence $(t_j, t_j k_j + d_j)$, compactness bounds k_j , and therefore d_j is bounded; take convergent subsequences of k_j and d_j . Its height-one section is $K + D$, so one implication follows by taking an affine section.

Conversely, homogenize a lift (4) of $S = K + D$ to

$$t \geq 0, \quad tA_0 + \sum_i y_i A_i + \sum_j z_j B_j \succeq 0.$$

At $t > 0$ the visible point is precisely (t, y) with $y \in tS$. If $(0, y, z)$ is feasible, add any nonnegative multiple of it to a feasible lift $(1, s, z_s)$. This shows $s + \alpha y \in S$ for every $\alpha \geq 0$. Consequently y is a recession direction of S . The recession cone of $K + D$ is D : writing $s + \alpha y = k_\alpha + d_\alpha$ and dividing by $\alpha \rightarrow \infty$ proves $y \in D$, while the opposite inclusion is immediate. Thus the homogenized lift has no height-zero direction outside D . Its projection may omit some directions in D ; adding the polyhedral cone $\{0\} \times D$ fills them. At positive height this addition leaves tS unchanged because $S + D = S$. The resulting projection is exactly (35). $\square$

Apply the lemma to $\mathcal{Q}_n$ and $\mathcal{H}_n = \mathcal{Q}_n + D_n$, where $D_n = \{(0, \text{diag}(s)) : s \geq 0\}$. Under the evaluation pairing for quadratics,

$$(\widehat{\mathcal{Q}_n})^* = \mathcal{P}_n, \quad (\widehat{\mathcal{H}_n})^* = \mathcal{P}_n^+. \quad (36)$$

Indeed, evaluation on $(1, x, xx^\top)$ requires nonnegativity on C_n , and evaluation on the additional diagonal rays requires $b_i \geq 0$. This identifies both dual cones without discarding the constant coordinate or the recession directions.

6.2 Finite lifts through dimension three

For completeness, we give the established construction that supplies the positive side of the threshold [1]. Let $\text{CP}_r = \text{cone}\{vv^\top : v \in \mathbb{R}_+^r\}$ and $\text{DNN}_r = \{W \in \mathbb{S}^r : W \succeq 0, W \geq 0 \text{ entrywise}\}$. The classical identity $\text{CP}_r = \text{DNN}_r$ holds for $r \leq 4$ [8, 12].

Triangulate C_n into its $n!$ order simplices. If V_π is the matrix of the $n + 1$ vertices of simplex π , introduce $W_\pi \in \text{DNN}_{n+1}$ and impose

$$\sum_\pi \mathbf{1}^\top W_\pi \mathbf{1} = 1, \quad m = \sum_\pi V_\pi W_\pi \mathbf{1}, \quad Y = \sum_\pi V_\pi W_\pi V_\pi^\top. \quad (37)$$

For $n \leq 3$ this is an exact lift of $\mathcal{Q}_n$. To verify exactness, decompose $W_\pi = \sum_\ell v_\ell v_\ell^\top$ with $v_\ell \geq 0$. Each nonzero summand represents the simplex point $V_\pi v_\ell / (\mathbf{1}^\top v_\ell)$ with mass $(\mathbf{1}^\top v_\ell)^2$. These masses sum to one in (37). Conversely, each simplex atom has such a rank-one representation. Adding D_n gives a lift of $\mathcal{H}_n$. The same argument applies to the quadratic moment hull of every polytope of dimension at most three, after triangulation in its affine hull.

6.3 The four-variable obstruction

The proof uses completely positive maps on real closed fields, following Bodirsky et al. [3]. Appendix C states the precise external inputs and explains why such maps preserve a finite semidefinite lift. The needed consequence is as follows. Let H be the Horn matrix, with diagonal entries 1, entries -1 on the cycle $12, 23, 34, 45, 51$, and entries 1 at the remaining off-diagonal positions. There are a real closed field $F \supseteq \mathbb{R}$ and a function $f : \mathbb{Q}^5 \rightarrow F$ such that $f(0) = 1$,

$$\eta := \sum_{i,j=1}^5 H_{ij} f(2e_i + 2e_j) < 0, \quad (38)$$

and the coefficient map

$$L_f \left(\sum_a c_a \varepsilon^a \right) = \sum_a f(a) c_a \varepsilon^a \quad (39)$$

is unital and completely positive on the real closed Hahn field $E = F((\varepsilon^{\mathbb{Q}^5}))$. Order $\mathbb{Q}^5$ lexicographically, with its first coordinate most significant. A monomial with positive leading exponent is then smaller than every positive real constant.

Theorem 6.2. *The sets $\mathcal{Q}_n$ and $\mathcal{H}_n$, and the cones $\mathcal{P}_n$ and $\mathcal{P}_n^+$, are spectrahedral shadows if and only if $n \leq 3$.*

Proof. The preceding construction, Lemma 6.1, and closed-cone duality prove the assertion for $n \leq 3$.

For $n = 4$, put $u_1 = 1$ and $u_{k+1} = x_k$, and consider the quadratic over E

$$q(x) = \sum_{i,j=1}^5 H_{ij} \varepsilon^{2e_i+2e_j} u_i u_j. \quad (40)$$

The Horn matrix is copositive. This remains true over E by transfer for real closed fields, so $q(x) \geq 0$ for $x \in C_4(E)$. Its square coefficients are $\varepsilon^{4e_2}, \dots, \varepsilon^{4e_5} > 0$, and hence $q \in \mathcal{P}_4^+(E)$.

Apply L_f to the coefficients of q and evaluate at

$$\xi_k = \varepsilon^{2e_1-2e_{k+1}}, \quad k = 1, \dots, 4.$$

Every ξ_k is positive and smaller than 1 by the chosen lexicographic order. Moreover $\varepsilon^{2e_i} u_i(\xi) = \varepsilon^{2e_1}$ for all i . Therefore

$$L_f(q)(\xi) = \varepsilon^{4e_1} \sum_{i,j=1}^5 H_{ij} f(2e_i + 2e_j) = \varepsilon^{4e_1} \eta < 0. \quad (41)$$

If either $\mathcal{P}_4$ or $\mathcal{P}_4^+$ had a finite semidefinite lift, the same lift would describe that cone over E , and the unital completely positive map L_f would preserve it. Equation (41) contradicts this preservation. Neither cone is a shadow.

For $n > 4$, restrict to quadratics depending only on $x_1, \dots, x_4$. This linear section of $\mathcal{P}_n$, respectively $\mathcal{P}_n^+$, is $\mathcal{P}_4$, respectively $\mathcal{P}_4^+$, so the larger cones cannot be shadows. Finally, (36) and Lemma 6.1 rule out lifts of $\mathcal{Q}_n$ and $\mathcal{H}_n$ for all $n \geq 4$. $\square$

The constant coordinate in (40) supplies the fifth copositive variable. The evaluation point places the resulting obstruction in an infinitesimal neighborhood of a cube vertex. The failure of $\text{CP}_5 = \text{DNN}_5$ alone would only rule out the construction (37); Theorem 6.2 rules out every

finite semidefinite construction. This conclusion is consistent with exact separation methods for low-dimensional box cones [5]: a separation procedure need not give a finite lift. The theorem concerns a single exact formulation of the joint moment hull. Particular objective epigraphs may have simpler lifts, and convergent semidefinite hierarchies remain possible. At fixed dimension, the existence of an exact separation algorithm and the absence of a finite lift are compatible statements; the theorem makes no algorithmic hardness claim.

6.4 Simple vertices of other polytopes

The argument depends on a local orthant at one vertex. The following form states the geometric condition directly. For a closed convex semialgebraic cone $K \subseteq \mathbb{R}^m$, define

$$\text{COP}(K) = \{M \in \mathbb{S}^m : z^\top M z \geq 0 \text{ for every } z \in K\}, \quad \text{CP}(K) = \text{cl cone}\{zz^\top : z \in K\}.$$

Theorem 6.3. *Suppose that a linear map $\ell = (\ell_1, \dots, \ell_5) : \mathbb{R}^m \rightarrow \mathbb{R}^5$ and a real $\delta > 0$ satisfy*

$$\ell(K) \subseteq \mathbb{R}_+^5, \quad \{u \geq 0 : u_k \leq \delta u_1 \ (k = 2, \dots, 5)\} \subseteq \ell(K). \quad (42)$$

Then neither $\text{COP}(K)$ nor $\text{CP}(K)$ is a spectrahedral shadow.

Proof. Use the fields and map in (38)–(39). The form

$$M(z) = \sum_{i,j=1}^5 H_{ij} \varepsilon^{2e_i+2e_j} \ell_i(z) \ell_j(z)$$

is nonnegative on $K(E)$, since $\ell(z) \geq 0$ there. The inclusion in (42) is a first-order statement with real parameters and therefore remains valid over E . In particular there is $w \in K(E)$ with

$$\ell(w) = (1, \varepsilon^{2e_1-2e_2}, \dots, \varepsilon^{2e_1-2e_5}).$$

Because L_f is real-linear, it acts termwise on the real coefficients of $\ell_i \ell_j$. Consequently $L_f(M)(w) = \varepsilon^{4e_1} \eta < 0$. Preservation under L_f contradicts a finite lift of $\text{COP}(K)$. The identity $\text{CP}(K)^* = \text{COP}(K)$ and closed-cone duality give the second assertion. $\square$

Corollary 6.4. *Let K be a full-dimensional pointed polyhedral cone of dimension at least five with a simple extreme ray, meaning a ray contained in exactly $\dim K - 1$ facets. Then $\text{COP}(K)$ and $\text{CP}(K)$ are not spectrahedral shadows. If a polytope X has dimension at least four and a simple vertex, then its cone of nonnegative quadratics and its quadratic moment hull have no finite semidefinite lifts.*

Proof. At a simple extreme ray, independent incident facet forms give nonnegative coordinates transverse to the ray. A form strictly positive on the cone gives the first coordinate. Sufficiently small nonnegative transverse coordinates satisfy the other, strictly slack facet inequalities. This proves (42); Appendix C.2 gives the uniform bound and covers height zero.

For a polytope X , take the cone over $\{1\} \times X$ in its affine span. It is pointed, and the ray through $(1, v)$ is simple whenever v is a simple vertex. Homogeneous quadratic forms on this cone correspond to quadratics on X . The cone is nonnegative exactly when the corresponding quadratic is nonnegative on X . The moment-hull assertion follows from compact homogenization and duality as in (36). $\square$

Every vertex of the cube is simple. The corollary does not give a classification of polytopes without a simple vertex.

6.5 Proper faces of the four-variable hull

The obstruction in Theorem 6.2 cannot be exhibited by finding a proper face of $\mathcal{Q}_4$ without a finite lift.

Proposition 6.5. *Every proper face of $\mathcal{Q}_4$ is a spectrahedral shadow.*

Proof. An exposed proper face is the moment hull of the cube zero set of a nonzero quadratic $p \geq 0$. On the relative interior of a cube face containing a zero, the restriction of p has zero gradient and a positive semidefinite Hessian. Its zero set on that face is therefore the intersection of the face with an affine subspace. These intersections form a finite union of polytopes of dimension at most three: cube boundary faces already have that dimension, and a nonzero quadratic cannot vanish on a four-dimensional open set. Their moment hulls have finite lifts by triangulation. The convex hull of finitely many compact sets with finite lifts also has a finite lift, using their closed homogenized cones and weights summing to one. Thus every exposed proper face has a finite lift.

For a general proper face, take a supporting hyperplane at a point in its relative interior. The resulting exposed face contains it. Repeating this operation inside the containing face reaches the given face after finitely many dimension drops. Affine sections preserve finite lifts. Appendix C.3 supplies the zero-set and face-chain details. $\square$

7 Sparse quadratic hulls and graph minors

The dense four-variable obstruction extends to sparse models through graph minors. The distinction between positive and negative diagonal coordinates is essential. A nonnegative square coefficient pairs with a diagonal epigraph, whereas a nonpositive square coefficient pairs with a diagonal hypograph.

Let $G = (V, E, L_+, L_-)$, where E is a set of unordered pairs of distinct vertices and $L_+, L_- \subseteq V$ are disjoint. Put $P = L_+$ and $R = V \setminus P$. The retained coordinates are m_i for $i \in V$, Y_{ij} for $ij \in E$, and Y_{ii} for $i \in L_+ \cup L_-$. Define the compact graph moment hull

$$\mathcal{K}(G) = \text{conv}\{(x, (x_i x_j)_{ij \in E}, (x_i^2)_{i \in L_+ \cup L_-}) : x \in [0, 1]^V\}.$$

Let $D(G)$ add nonnegative slack on positive diagonals and nonpositive slack on negative diagonals, with zero entries on all other retained coordinates. The signed hull is

$$\mathcal{H}(G) = \mathcal{K}(G) + D(G). \quad (43)$$

Equivalently, it is the convex hull of points with exact edge products, $Y_{ii} \geq x_i^2$ for $i \in L_+$, and $Y_{ii} \leq x_i^2$ for $i \in L_-$. It is closed because $\mathcal{K}(G)$ is compact and $D(G)$ is closed. This is the graph hull studied by Khajavirad [11].

7.1 A K_4 minor among positive vertices

Theorem 7.1. *If the graph $G[P]$ induced by the positive-loop vertices contains a K_4 minor, then $\mathcal{H}(G)$ has no finite semidefinite lift.*

Proof. Choose disjoint connected branch sets $C_1, \dots, C_4 \subseteq P$ witnessing the minor. In each nonsingleton branch set choose a spanning tree, and choose one edge e_{ab} between C_a and C_b for every $a < b$. The affine function

$$F = \sum_{ij \text{ in the chosen trees}} (Y_{ii} + Y_{jj} - 2Y_{ij}) \quad (44)$$

is nonnegative on $\mathcal{H}(G)$. Indeed, each summand at a generating point is $(x_i - x_j)^2 + s_i + s_j$, with $s_i, s_j \geq 0$.

On the face $F = 0$, every positive-weight atom is constant on each branch set. All diagonal slack on nonsingleton branch sets vanishes, since each vertex belongs to a tree edge. Slack at singleton branch sets may remain. Choose a representative $r_a \in C_a$ and project this face onto

$$u_a = m_{r_a}, \quad Z_{aa} = Y_{r_a r_a}, \quad Z_{ab} = Y_{e_{ab}} \quad (a < b).$$

The image is exactly $\mathcal{Q}_4 + D_S$, where D_S is nonnegative diagonal slack on the singleton branch sets. For the forward inclusion, the common branch values of each atom give a cube atom in four variables, and only those singleton slacks survive. For the reverse inclusion, assign the coordinate u_a of any cube atom to every vertex in C_a , assign zero outside the branch sets, and take exact squares and products in the source graph. Add any desired singleton slack and take mixtures.

If $\mathcal{H}(G)$ had a finite lift, so would its face and this projection. Adding the remaining nonnegative diagonal rays would then give a finite lift of $\mathcal{H}_4$, contradicting Theorem 6.2. $\square$

Extra edges, extra vertices, and negative-loop vertices do not affect the proof. The branch sets must lie in P : a negative or absent diagonal cannot provide the nonnegative squared-difference expression in (44). The theorem therefore gives an obstruction in the positive-induced graph, rather than in the full interaction graph. It applies, for example, to every subdivision of K_4 with positive loops at all subdivision vertices.

K_4 already prevents a finite lift, although every proper face of its compact moment hull has one by Proposition 6.5.

7.2 Minor closure for all-positive graphs

When every vertex has a positive loop, write $\mathcal{H}(G)$ for this all-positive hull.

Proposition 7.2. *If $\mathcal{H}(G)$ has a finite semidefinite lift and G' is a minor of G , then $\mathcal{H}(G')$ has a finite semidefinite lift.*

Proof. Deleting an edge or a vertex is a coordinate projection. To contract an edge ij , impose the exposed-face equation $Y_{ii} + Y_{jj} - 2Y_{ij} = 0$. As in (44), each atom then satisfies $x_i = x_j$, and the two diagonal slacks vanish. Keep one first-moment coordinate and one diagonal coordinate for the merged vertex. For an edge from the merged vertex to a neighbor k , keep either Y_{ik} or Y_{jk} that is present; if both are present, they agree on this face. Keep all other coordinates. The projection has the graph moment hull of G/ij , with nonnegative diagonal slack at every vertex except the merged vertex. Add the diagonal ray at that vertex to obtain exactly $\mathcal{H}(G/ij)$. All three operations preserve finite lifts. $\square$

The diagonal-ray addition is necessary. The contraction face forces the merged diagonal to be an exact second moment; merely projecting that face does not recover its full positive diagonal epigraph.

7.3 Finite lifts for small positive components

At the other end, components with at most three positive vertices admit finite lifts. This combines binary rounding outside P , the established three-variable moment lift, and consistent binary marginal tables. We give the formulation because it makes clear which graph width controls its size.

For each connected component C of $G[P]$, let $B_C = N(C) \cap R$. Form a graph T on R by keeping its original edges and making every B_C a clique. We call T the *torso* after the positive components are removed.

Theorem 7.3. *If every connected component of $G[P]$ has at most three vertices, then $\mathcal{H}(G)$ has a finite semidefinite lift whose matrix blocks have order at most four.*

More precisely, suppose a tree decomposition of T with bags $\mathcal{B}$ and width τ is supplied. The construction uses $\sum_{B \in \mathcal{B}} 2^{|B|}$ nonnegative binary table entries and

$$\sum_C |C|! 2^{|B_C|} \quad (45)$$

positive semidefinite blocks of order $|C| + 1$, each constrained to be entrywise nonnegative. Apart from the original retained coordinates, the number of variables and affine constraints is

$$O\left(|\mathcal{B}|2^{\tau+1} + \sum_C |C|! 2^{|B_C|}\right).$$

This count does not count every nonzero coefficient in the displayed linear maps.

Proof. First make the coordinates in R binary. At an original generating point, independently round x_j to a Bernoulli variable with mean x_j for every $j \in R$, leaving coordinates in P fixed. The first moments and every edge product remain unchanged. A negative-loop coordinate can be restored by downward slack, because the rounded square has mean $x_j \geq x_j^2$. Positive diagonal coordinates are unchanged. Thus the original generator lies in the convex hull of generators with binary R . Conversely, those binary generators are allowed original generators. It is enough to describe their mixtures.

For each bag B of the decomposition introduce a nonnegative table $p_B : \{0, 1\}^B \rightarrow \mathbb{R}_+$ with total mass one. Require adjacent bags to agree on the marginal table of their intersection. These tables admit a joint binary law on R . To construct it, root the decomposition tree and attach each child table by its conditional law given the separator state. Separator states of zero mass never occur and may be assigned arbitrary conditional laws. The running-intersection property ensures that the joint law has every prescribed bag marginal. This construction is finite. If R is empty, use a single empty bag whose sole table entry is 1; the empty boundary assignment then has probability 1.

Each B_C is a clique in T , so it is contained in a bag. For every $b \in \{0, 1\}^{B_C}$ introduce the homogenized simplex lift of the component cube $[0, 1]^{B_C}$ with total mass equal to the marginal probability of b . Writing $k = |C|$ and V_π for an order-simplex vertex matrix, its variables satisfy

$$\begin{aligned} W_{C,b,\pi} &\succeq 0, & W_{C,b,\pi} &\geq 0, \\ \sum_\pi \mathbf{1}^\top W_{C,b,\pi} \mathbf{1} &= \Pr(X_{B_C} = b), \\ m_{C,b} &= \sum_\pi V_\pi W_{C,b,\pi} \mathbf{1}, & Z_{C,b} &= \sum_\pi V_\pi W_{C,b,\pi} V_\pi^\top. \end{aligned}$$

Because $k + 1 \leq 4$, $\text{CP}_{k+1} = \text{DNN}_{k+1}$ proves exactness just as in (37). Zero mass forces every entry of every W to be zero, so no conditional moments can remain at a zero-probability state.

Sum $m_{C,b}$ and $Z_{C,b}$ over b for component first and second moments. For an edge ij with $i \in C$ and $j \in B_C$, use

$$Y_{ij} = \sum_b b_j (m_{C,b})_i.$$

Moments on R come from the bag tables. At a negative-loop vertex j , set $Y_{jj} = m_j - t_j$ with $t_j \geq 0$; at a positive-loop vertex add nonnegative diagonal slack to the component second moment.

Every original generator and every mixture gives feasible variables. Conversely, begin with the joint binary law on R just constructed. Given the full binary assignment, sample each continuous component from the law specified by its boundary assignment b , independently across components. The simplex decomposition supplies these component laws. There are no edges between distinct positive components. Hence this joint law realizes every retained first moment, edge product, and unslacked diagonal. Adding the stated signed slacks proves reverse inclusion.

The table count is immediate. Each pair (C, b) has $k!$ simplex blocks, each of bounded order. Separator equalities use at most $O(|\mathcal{B}|2^{\tau+1})$ entries, and attachment equalities use one equation per boundary state. This proves the stated count. $\square$

For finite representability alone one may take a single bag containing all of R ; the formulation can then be exponentially large. A small formulation follows when the supplied torso decomposition and the sum in (45) give a small total size. Original graph treewidth alone does not control this count: completing component boundaries to cliques can increase torso width. The theorem also concerns quadratic graph moments; it does not describe a retained product of three continuous coordinates.

Theorem 7.3 extends the two-positive-vertex component construction of Khajavirad [11] by using the known three-variable simplex lift. Its sufficiency and Theorem 7.1 leave an explicit gap: components with at least four vertices and no K_4 minor are not classified here.

7.4 The remaining path and star question

Every connected graph with at least four vertices contains a four-vertex path P_4 or a three-leaf star $K_{1,3}$ as a subgraph. Indeed, a spanning tree either has a vertex of degree at least three or is a path. Together with Proposition 7.2 and Theorem 7.3, this gives a precise reduction: if both $\mathcal{H}(P_4)$ and $\mathcal{H}(K_{1,3})$ lack finite lifts, then an all-positive graph has a finite lift exactly when every component has at most three vertices. We do not assume either missing result.

There is also an exact reduction of the forest question to submodular quadratics. Let $\mathcal{P}_G^+$ be the cone of quadratics nonnegative on $[0, 1]^V$, with square coefficients nonnegative and cross coefficients zero outside E . Let

$$\mathcal{S}_-(G) = \{q \in \mathcal{P}_G^+ : c_{ij} \leq 0 \text{ for } ij \in E\}.$$

These are the submodular quadratics in the sparse cone.

Proposition 7.4. *For a forest G , the hull $\mathcal{H}(G)$ has a finite semidefinite lift if and only if $\mathcal{S}_-(G)$ is a spectrahedral shadow.*

Proof. Complementing a coordinate, $x_i \mapsto 1 - x_i$, preserves box nonnegativity and every square coefficient. It reverses the sign of each incident cross coefficient. On a forest the complementations can be chosen independently along edges: root each tree, choose the root sign, and then choose each child sign to make its parent-edge coefficient nonpositive. Therefore every member of $\mathcal{P}_G^+$ is the complementation image of a member of $\mathcal{S}_-(G)$. Each such image remains in $\mathcal{P}_G^+$, so

$$\mathcal{P}_G^+ = \sum_{\sigma} \sigma(\mathcal{S}_-(G)),$$

where the finite sum ranges over coordinate complementations, acting linearly and invertibly on polynomial coefficients. If $\mathcal{S}_-(G)$ is a shadow, the sum is a shadow. Conversely, $\mathcal{S}_-(G)$ is obtained from $\mathcal{P}_G^+$ by linear inequalities on the cross coefficients and therefore is a shadow whenever $\mathcal{P}_G^+$ is.

Lemma 6.1 and closed-cone duality identify finite representability of $\mathcal{P}_G^+$ with that of $\mathcal{H}(G)$, proving the assertion. $\square$

In particular, a finite lift for the submodular part of the dense four-variable positive cone would yield lifts for both P_4 and $K_{1,3}$ by setting the other cross coefficients to zero. The dense nonrepresentability theorem does not decide this sign-restricted question. Nor does failure of a particular relaxation decide it: the issue is the existence of an arbitrary finite extended formulation.

8 Numerical comparisons

The compact family block is useful if a relaxation has a gap that can be removed using three-variable moments. This section asks two separate questions: whether the block closes such gaps, and whether the tested instance classes exhibit them. The results support the first claim on constructed instances. They give little evidence for an additional gain on the benchmark and random instances. All experiments concern root relaxations; no branch-and-bound implementation was tested.

8.1 Models and comparison protocol

We minimize $x^T H x + g^T x$ over C_n , with lifted objective $f(m, Y) = \langle H, Y \rangle + g^T m$. For $\bar{u} = (1, u^T)^T$, we use $\mathcal{M}_3 = \text{conv}\{\bar{u}\bar{u}^T : u \in C_3\}$ for the normalized 4×4 matrix form of $\mathcal{Q}_3$. The baseline, denoted B , contains the Shor condition, McCormick inequalities, diagonal caps $Y_{ii} \leq m_i$, and the four triangle inequalities on each considered triple. For dense instances the Shor matrix has order $n + 1$ and all pairs are recorded. For sparse instances it is imposed on maximal cliques of a chordal pattern containing the objective support. The other constraints then use pattern pairs and triples within those cliques. The comparisons introduce no first or second moment outside this pattern.

The strengthening methods are as follows.

- K The three-positive-variable disjoint-support system of Khajavirad [11, eq. (17)]: eight level-three RLT inequalities, twelve order-two blocks, and six order-three PSD blocks per selected triple. Its order-four block is already in the Shor condition. Higher monomials are shared when the same monomial occurs in different triples. Larger positive-variable sets and nonempty negative-variable sets were not tested.
- A The eight trilinear inequalities and the switched and permuted second-order-cone constraints of Anstreicher and Puges [2, eqs. (14)–(16)].
- F The family block of Section 4, added only for a violated triple and orientation. The 24 orientations choose the distinguished coordinate and independently complement the three coordinates.
- X The exact three-variable moment lift (60), using five tetrahedra. A selected triple adds five 4×4 DNN blocks and ten linking equations.

Combined labels mean that the corresponding systems are imposed together. In particular, KA adds both existing systems, and KAF adds the family blocks to KA . The method XF uses the exact lift but selects triples by family separation. Individual family cuts and individual cube-hull cuts were also compared with the blocks.

Family separation minimizes $v^T(B - bb^T)v$ over $v \geq 0$, $\mathbf{1}^T v = 1$, by the fifteen-support calculation in Proposition D.1. For K , A , KA , and X , selection instead uses the full hull-depth problem (61). Hull depth is the smallest evaluation at the triple moments of a cube-nonnegative quadratic with

$\int_{C_3} q(u) du = 1$; it is negative exactly outside $\mathcal{M}_3$ in exact arithmetic. This is a stronger test than separation of either existing inequality system, and its cost is reported as separation time. Triples with a coordinate within 10^{-6} of a box bound are skipped by this hull-based rule. At an exact bound, Shor and McCormick reduce the triple to a two-variable problem with an exact baseline [1, Theorem 6]; the numerical skip is a heuristic extension of that fact.

The usual violation threshold is 10^{-6} . Triangles are separated again after each solve. The default limits are 200 new blocks or triples per round, 25 rounds, and 3600 seconds per method; the 3000-triangle chains use 1000 additions and 15 rounds, and the benchmark comparisons use 10 rounds. A run can also stop when the relative bound change is below 10^{-7} over three rounds. These limits define selective comparisons; they are not full-system optimality certificates.

The main solver is Clarabel 0.11.1 [7, 9] with gap and feasibility tolerances 10^{-8} and one thread. The three-variable comparisons use 10^{-10} ; the hull-depth subproblems use 10^{-9} . SCS 3.3.1 [15, 16] provides selected accuracy comparisons. Reported relaxation values use the dual-residual correction in Proposition D.4. We denote these floating-point lower-bound estimates by L_M for method M . They are not interval certificates. Feasible objective values U come from published benchmark optima, small-instance face enumeration, or the best archived Gurobi and local-search incumbents. When U is only a feasible value, the fraction

$$\frac{L_M - L_B}{U - L_B} \quad (46)$$

is a conservative estimate of gap closure, conditional on the accuracy of the reported lower bounds. It is not a claim that U is optimal.

8.2 Three-variable gaps

To isolate the effect of a triple block, 109 objectives were constructed from separating quadratics of points of B outside $\mathcal{M}_3$. The separating quadratics satisfy $\int_{C_3} q(u) du = 1$ and have positive coefficients of all three square terms. Their baseline gaps have median 0.072 and range 0.0069–0.123. The exact lift gives the reference value. All systems are imposed in full for this comparison, including all 24 family orientations.

Table 1: Fractions of the three-variable gap closed on 109 constructed objectives, computed as $(L_M - L_B)/(L_X - L_B)$.

Method	Mean	Median	Minimum
K	0.9084	0.9177	0.7668
A	0.5766	0.5811	0.2619
KA	0.9084	0.9177	0.7668
F	1.0000	1.0000	0.999983
KAF	1.0000	1.0000	0.999912

Every F primal value is at most 5.5×10^{-9} below the corrected exact-lift reference. The small losses in its corrected-bound closures in Table 1 also reflect dual-residual correction; they are not accuracy guarantees at the solver’s requested tolerance. The existing systems leave a visible fraction of the gap. In separate selective runs, exactly one family orientation was violated at each baseline point, and its block closed the gap in one added-block solve. Individual family cuts had median closure 0.38 after one round and 0.9933 after 29 rounds; none met the stopping test within the

30-round limit. The comparison illustrates the difference between imposing the whole parameterized family and adding its current most violated member.

Additional searches did not find a gap between the full family relaxation and the exact lift. They included 2180 perturbations of the constructed objectives and 256 556 sampled baseline points satisfying every family orientation. The smallest computed hull depth in the latter set was -6.4×10^{-8} . This evidence is consistent with the three-variable completeness question, but does not prove it. The samples are finite, nonuniform, and assessed in floating point.

8.3 Constructed sparse instances

The same hard objectives were assembled into three sparse families. A chain has m disjoint triangles, with consecutive triangles joined by one bridge edge; the bridge scale is denoted η . A cactus adds each new triangle by sharing one vertex with an earlier triangle. The third family puts hard objectives on the triangles of a 2-tree or 3-tree, so triangles share edges. Random weights, permutations, and coefficient perturbations prevent exact repetition of a single subproblem. These constructions test the family when local gaps are deliberately present; they do not model an established application class.

Table 2: Representative constructed instances, all with seed 1. Closures use (46). Solve times are cumulative seconds after the common baseline solve. F/X counts family blocks and lifted triples, respectively. In the last two columns, each slash separates the F value from the X value; it is not a ratio.

Instance	n	KA	F	X	Solve F/X	Blocks F/X
Chain, $m = 100, \eta = 0$	300	.9956	1.0000	1.0000	.47/2.17	78/78
Chain, $m = 300, \eta = .3$	900	.8577	.8608	.8597	2.25/24.62	110/223
Chain, $m = 1000, \eta = .3$	3000	.8985	.9021	.8950	8.13/110.62	346/997
Chain, $m = 3000, \eta = .3$	9000	.8847	.8889	.8807	26.20/201.36	1051/2999
Cactus, $m = 30$	61	.8345	.8718	.8718	.04/.14	27/27
Cactus, $m = 300$	601	.7440	.7501	.7489	2.13/11.71	235/284
Cactus, $m = 1000$	2001	.7135	.7208	.7181	19.37/97.19	805/997
Hard 2-tree, $n = 60$	60	.0046	.0051	.0041	.06/.35	4/17

On the 16 chains and eight cacti, the F and X primal values differ by at most 5.6×10^{-6} relative. The larger differences between their corrected closures in Table 2 reflect the accuracy limitations discussed in Section 8.5; selected higher-accuracy comparisons support a numerical tie. The family added 0.31–3.73 percentage points of closure beyond KA when KA had not already matched the strengthened values. Combining K and A with the family gave no detectable further gain. With $\eta = 0$, the chain separates into independent three-variable problems, so the triple-level bound is the true optimum. Bridges and vertex sharing can leave a global gap after all tested triple strengthenings.

The block sizes explain one possible advantage: a family orientation adds one 5×5 PSD block and six nonnegative auxiliaries, whereas the exact lift adds five 4×4 blocks, thirty off-diagonal nonnegativity constraints, and ten equations. If all 24 orientations are needed, the family can be larger. In the selective runs, however, 7598 violated orientation–triple pairs occurred on 7459 violated triples, or 1.02 orientations per triple on average.

Selection also matters. At $m = 3000$, X lifts 2999 triples, whereas F adds 1051 blocks. The method XF uses the same family selection rule as F , although its different iterates can select different triples. On chains, its solve times range from 0.23 to 2.05 times those of F , in separate

runs. Thus the chain comparison does not isolate an advantage of the block type. On cacti, XF and F were run in the same invocation: XF took 2.1–5.7 times the solve time of F . This supports a block-size advantage there, subject to the timing limitations below. Full records, including separation time and model size, are provided in the companion archive.

Edge overlap gave a different outcome. Across eight hard k -tree instances, the methods closed at most 0.514% of $U - L_B$, and the hull test selected only zero to seven triples at the baseline points. Summing individually hard three-variable objectives therefore does not ensure that the combined problem retains substantial three-variable gaps. This is an empirical observation for these instances, not a theorem about overlapping triangles.

8.4 Benchmark and random instances

The benchmark uses the BoxQP collection [4]. The collection uses the maximization objective $\frac{1}{2}x^T Qx + c^T x$; we set $H = -Q/2$ and $g = -c$ and negate its reference optima to obtain the minimization form used here. Table 3 summarizes the tested classes. The sparse positive-diagonal instances are random 2-trees and 3-trees with positive integer square coefficients and signed integer edge and linear coefficients. Their clique triples all have three positive square coefficients.

Table 3: Benchmark and random-instance findings. “Tight” refers to the stated numerical tolerance, not an exact certificate.

Class	Count	Finding
BoxQP benchmark	99	B has relative gap below 10^{-6} on 82 instances. No family orientation was selected in the 17 original gap audits.
Small dense, BoxQP-style generator	18 000	$n = 5, \dots, 10$; all baseline gaps below 2.3×10^{-7} relative using corrected dual estimates.
Small dense, AP variant	12 000	One gap of 3.6×10^{-6} relative, already closed by KA ; no additional family gain.
Sparse, positive diagonals	16	$n = 100$ –3000; no family orientation selected. Computed minimum triple depths at least -3×10^{-7} .

The original BoxQP audits illustrate why a numerical hull violation must be checked against residuals of the baseline itself. They evaluated every triple in each of the 17 instances with a recorded gap. Four computed depths were below -10^{-6} , but these were explained by triangle inequalities left violated by the original early stopping rule. A triangle quadratic has uniform mean $1/4$; a violation r therefore forces normalized hull depth at most $-4r$. Likewise, a diagonal-cap violation $Y_{ii} - m_i = r > 0$ forces depth at most $-6r$, since $x_i - x_i^2$ has uniform mean $1/6$. These are errors in satisfying B , not evidence of a gap between a feasible baseline point and the triple hull.

One complete stricter audit is available, for **spar090-075-1**. It separates triangles at 10^{-8} and evaluates all 117 480 triples. The computed minimum depth is -8.10×10^{-9} , with reported primal infeasibility 2.02×10^{-9} . This computed depth is not a one-sided certificate: a remaining triangle residual implies that the exact minimum is below -1.4×10^{-8} , and stored depths exceed exact triangle or cap upper bounds by up to 9.67×10^{-8} . The gain expression in (64), evaluated at the computed depths, is 0.045% of the recorded gap, or 0.54% after adding the primal-to-dual margin. If every computed depth overestimates its exact value by at most 10^{-7} , these figures become at most 0.60% and 1.10%, respectively. That assumption and exact baseline feasibility are not certified. Only this one strict all-triple audit was completed; its estimate does not apply uniformly to the other 16 instances. The remaining evidence is consistent with small triple-local gains, but does not prove a uniform bound.

Proposition D.3 does not cover K when higher moments are shared across triples. A separate test on `spar050-050-1` added K to all 19 600 triples. Its primal value changed by 4×10^{-7} against a gap of 1.72. This is a negative finding on one instance, not a general exclusion of gains from higher-order consistency.

The small dense BoxQP-style generator reimplements the published distribution with different random numbers. An additional generator was based on the description in Anstreicher and Puges [2], but includes the diagonal in the density mask. It does not reproduce their twelve reported gap instances. Its one gap, at $n = 9$, is approximately 0.00104 in absolute value. A targeted audit finds two violated triples; K , A , KA , and two family blocks all close the gap to their observed accuracy, while KAF adds no family block after KA . Consequently, the study supplies a natural-generator triple-level gap, but no natural instance on which the family adds a detectable gain beyond the existing systems. These generator-specific findings do not justify a statement about all random box quadratic programs.

8.5 Accuracy, timing, and interpretation

The companion archive contains compact tabulations, constructed-instance data, selected raw records, implementation snapshots, and source checksums. It supports independent inspection of the reported tables. Reproduction of all benchmark audits and sampling campaigns also requires the larger original input and output archives and their software dependencies.

The largest models often returned **AlmostSolved**. The exact lift had the largest numerical degradation: each triple adds several blocks and equations linking small lifted matrices to the original moments. For example, on the 1000-triangle chain in Table 2, Clarabel reports corrected values -2502.697859 for F and -2502.791210 for X . At tolerance 10^{-6} , SCS instead gives -2502.697709 and -2502.697835 , respectively. Their relative difference is 5.05×10^{-8} , supporting a numerical tie. The lower Clarabel value for X is therefore an accuracy limitation, not evidence that the family is mathematically stronger than the exact lift. Primal-to-dual margins are retained in the records and must be considered before interpreting small differences.

Times are wall clock on a shared 36-core machine, whose load varied substantially. Methods ran sequentially, so even within-run ratios are affected by changing load. A separate quieter run of the 300-triangle chain reproduced the bounds but changed X/F from 10.9 to 7.5 and KA/F from 3.2 to 1.4. The tabulated solve times exclude the common baseline stage and separation time; they do not measure complete solver time. The model also stores the objective matrix densely, limiting the largest tested problem to 9000 variables. These observations support a useful compact strengthening on deliberately constructed triple gaps. They establish neither a general runtime advantage nor an improvement in a global optimization solver.

9 Implications and remaining questions

The three-variable results give a precise reason to strengthen a local relaxation. The five edge contacts of the new family cannot be matched by the allowed affine factors in a disjoint-support certificate. This failure persists for a strictly positive quadratic and is witnessed by rational moments that satisfy the specified recent SOC strengthening. Enforcing the family does not require a larger hierarchy of moments: one order-five block and six nonnegative scalars suffice for each orientation. The exact tetrahedral hull remains an alternative when complete local convexification is needed.

The closedness, rounding, and sign results narrow the remaining three-variable question. A missing extreme ray must have all three square coefficients positive and a positive product of mixed coefficients. Cube symmetries then reduce its sign pattern, and the contact arguments

exclude several zero configurations. Theorem 5.12 settles every sign pattern when all vertex values are positive: an extreme ray with positive square coefficients is an affine square or a symmetry copy of the strict family. Rays with a vertex zero still require classification beyond the boundary results proved here. The precise completeness question is stated in Section 5; none of the validity, enforcement, or representability results depends on a positive answer. The sampled moment and contact studies provide numerical evidence for the broader equality, rather than a proof.

The dimension threshold sets a different limit. In four or more variables, no finite collection of affine semidefinite constraints and auxiliary variables can represent the full quadratic moment hull, or its positive diagonal enlargement. The obstruction is local to a vertex and extends to other polytopes with a simple vertex. It does not prevent exact separation, convergent hierarchies, or finite formulations for particular objective classes. The proper-face result also shows that the obstruction can disappear after restricting to the zero set of any nonzero supporting quadratic in four variables.

For sparse models, local dimension alone is insufficient. Components with at most three positive vertices can be combined through binary boundary states to give a finite lift. A K_4 minor among positive vertices rules out any finite lift. The intervening graph class remains open, with the four-vertex path and the three-leaf star as the first cases identified here. The available finite construction can also be large: its size depends on the torso obtained by completing component boundaries, rather than only on the original graph width.

The computational evidence supports a limited practical conclusion. Selected family blocks can recover exact local-hull bounds on constructed instances designed around the missing inequalities. The tested natural instances give much less room for improvement, and the archived timings do not establish which formulation is fastest under matched selection and computing conditions. A solver implementation should therefore measure both the bound gained and the cost of each selected block. The exact separation and gain bounds in the appendices provide tools for that comparison, provided their arithmetic and locality hypotheses are respected.

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A Exact certificates for the disjoint-support gap

This appendix gives the complete rational certificate used in Proposition 3.2. All its entries are finite rational numbers, so positive definiteness and the SOC comparisons reduce to integer arithmetic.

A.1 Moments and localizing matrices

Table 4: The functional Λ on V . Each listed numerator is divided by 10000. The exponent (a, b, c) represents $x^a y^b z^c$.

Exponent	Numerator	Exponent	Numerator
(0, 0, 0)	10000	(1, 0, 0)	4041
(0, 0, 1)	1377	(1, 0, 1)	1025
(0, 0, 2)	681	(1, 0, 2)	629
(0, 1, 0)	4041	(1, 1, 0)	771
(0, 1, 1)	1025	(1, 1, 1)	674
(0, 1, 2)	629	(1, 1, 2)	590
(0, 2, 0)	3392	(1, 2, 0)	770
(0, 2, 1)	1222	(1, 2, 1)	598
(2, 0, 0)	3392	(2, 1, 0)	770
(2, 0, 1)	1222	(2, 1, 1)	598

For a disjoint pair (A, B) , let $u_{A,B}$ list 1 followed by the coordinates outside $A \cup B$, in increasing coordinate order. The localizing matrix is

$$M_{A,B} = \Lambda(w_{A,B} u_{A,B} u_{A,B}^T).$$

The following tables give every such matrix, with each matrix multiplied by 10000. Their entries follow directly by expanding the factors $1 - x_i$ against Table 4. The moments are invariant under exchanging x and y ; rows marked with multiplicity two represent the corresponding pair of matrices, with the free coordinates exchanged. This convention preserves the displayed matrix and the order of its basis.

Proposition A.1. *All 27 localizing matrices specified by Table 4 are positive definite. They consist of one order-four matrix, six order-three matrices, twelve order-two matrices, and eight scalar matrices.*

Proof. The unweighted matrix, with basis $(1, x, y, z)$, is

$$10000M_{\emptyset,\emptyset} = \begin{pmatrix} 10000 & 4041 & 4041 & 1377 \\ 4041 & 3392 & 771 & 1025 \\ 4041 & 771 & 3392 & 1025 \\ 1377 & 1025 & 1025 & 681 \end{pmatrix}. \quad (47)$$

Its four leading determinants, before dividing the matrix by 10000, are

$$10000, \quad 17590319, \quad 23512042198, \quad 45063248771.$$

For the order-three blocks, let u be either x or y , and let v be the other coordinate. The basis for a weight involving u is $(1, v, z)$; the basis for a weight involving z is $(1, x, y)$.

Weight	$10000M_{A,B}$	Multiplicity
$1 - u$	$\begin{pmatrix} 5959 & 3270 & 352 \\ 3270 & 2622 & 351 \\ 352 & 351 & 52 \end{pmatrix}$	2
u	$\begin{pmatrix} 4041 & 771 & 1025 \\ 771 & 770 & 674 \\ 1025 & 674 & 629 \end{pmatrix}$	2
$1 - z$	$\begin{pmatrix} 8623 & 3016 & 3016 \\ 3016 & 2170 & 97 \\ 3016 & 97 & 2170 \end{pmatrix}$	1
z	$\begin{pmatrix} 1377 & 1025 & 1025 \\ 1025 & 1222 & 674 \\ 1025 & 674 & 1222 \end{pmatrix}$	1

Their leading determinants are as follows; the order- r column must be divided by 10000^r to obtain the determinant of the unscaled block.

Weight	Order 1	Order 2	Order 3
$1 - u$	5959	4931598	5442129
u	4041	2517129	3854275
$1 - z$	8623	9615654	2810633517
z	1377	632069	279229016

For the order-two blocks, the basis is $(1, u)$ in the first four rows below, where $\{u, v\} = \{x, y\}$, and $(1, z)$ in the final three rows.

Weight	$10000M_{A,B}$	Multiplicity	Determinant numerator
$(1-v)(1-z)$	$\begin{pmatrix} 5607 & 2919 \\ 2919 & 1998 \end{pmatrix}$	2	2682225
$v(1-z)$	$\begin{pmatrix} 3016 & 97 \\ 97 & 172 \end{pmatrix}$	2	509343
$(1-v)z$	$\begin{pmatrix} 352 & 351 \\ 351 & 624 \end{pmatrix}$	2	96447
vz	$\begin{pmatrix} 1025 & 674 \\ 674 & 598 \end{pmatrix}$	2	158674
$(1-x)(1-y)$	$\begin{pmatrix} 2689 & 1 \\ 1 & 13 \end{pmatrix}$	1	34956
$x(1-y), (1-x)y$	$\begin{pmatrix} 3270 & 351 \\ 351 & 39 \end{pmatrix}$	2	4329
xy	$\begin{pmatrix} 771 & 674 \\ 674 & 590 \end{pmatrix}$	1	614

Each determinant numerator is divided by 10000^2 . Every first diagonal entry is positive. Finally, the eight scalar blocks, indexed by the cube vertex corresponding to their literal weight, are

Vertex	000	001	010, 100	011, 101	110	111
Numerator	2688	1	2919	351	97	674

Again all entries are divided by 10000. The multiplicities give all 27 blocks. Positivity of every displayed leading determinant proves the assertion by Sylvester's criterion. $\square$

For reference, the minimum leading principal determinant of each order, over every block where that order is present, is

$$\frac{1}{10000}, \quad \frac{307}{50000000}, \quad \frac{154171}{40000000000}, \quad \frac{45063248771}{10^{16}}. \quad (48)$$

The tables provide a positive-definiteness certificate without solving an SDP. The separating objective can likewise be reproduced directly:

$$10000\Lambda(p) = 2(3392) + 9(681) + 6(771) - 24(1025) - 2(4041) + 9(1377) + 2500 = -250.$$

A.2 All permuted and complemented SOC inequalities

For $s = (s_x, s_y, s_z) \in \{0, 1\}^3$, set $u_i = x_i$ when $s_i = 0$ and $u_i = 1 - x_i$ when $s_i = 1$. Write $Y_{ij}^s = \Lambda(u_i u_j)$ and $t_s = \Lambda(u_x u_y u_z)$. The entries needed for (12) are in the following table. As before, each entry is divided by 10000. Exchanging x and y gives the two omitted rows, 010 and 011.

s	Y_{xx}^s	Y_{yy}^s	Y_{zz}^s	Y_{xy}^s	Y_{xz}^s	Y_{yz}^s	t_s
000	3392	3392	681	771	1025	1025	674
100	5310	3392	681	3270	352	1025	351
001	3392	3392	7927	771	3016	3016	97
110	5310	5310	681	2689	352	352	1
101	5310	3392	7927	3270	5607	3016	2919
111	5310	5310	7927	2689	5607	5607	2688

Proposition A.2. *The functional in Table 4 satisfies all permuted and complemented inequalities in (12) strictly. The minimum slacks are 2831/4000000 for the first type and 124813/12500000 for the second type. All right-hand factors are strictly positive.*

Proof. For each complement pattern, form the three first-type slacks and the six second-type slacks

$$Y_{ii}^s Y_{jk}^s - t_s^2, \quad Y_{ii}^s (Y_{jj}^s + 3Y_{jk}^s) - (Y_{ij}^s + t_s)^2,$$

respectively. Direct substitution from the preceding table gives the following minima. The entries are integer numerators with denominator 10^8 .

Complement pattern	Minimum first-type slack	Minimum second-type slack
000	70775	998504
100, 010	1070783	3998528
001	6102308	35532766
110	1831208	8985128
101, 011	7494399	17444574
111	14090359	37232454

These eight patterns cover all 24 first-type inequalities and all 48 second-type inequalities. Their global minima reduce to the stated fractions. Every diagonal and cross moment in the table is strictly positive, which supplies the nonnegative factors required for the rotated SOC inequalities. The smallest such diagonal or cross factor is $352/10000 = 22/625$. $\square$

The same scalar blocks also verify the ordinary multilinear baseline. If λ_s denotes the scalar block at vertex s , then $\lambda_s > 0$ and $\sum_s \lambda_s = 1$. Marginalization gives all first and off-diagonal second moments. In particular, the off-diagonal RLT inequalities follow by summing suitable λ_s . The two ordinary triangle types follow from the identities

$$m_i + Y_{jk} - Y_{ij} - Y_{ik} = \Lambda(x_i(1-x_j)(1-x_k) + (1-x_i)x_jx_k),$$

$$1 - m_x - m_y - m_z + Y_{xy} + Y_{xz} + Y_{yz} = \Lambda((1-x)(1-y)(1-z) + xyz).$$

Together with the positive definite base moment matrix and the diagonal upper-bound slacks in Proposition 3.2, this verifies the claimed baseline for the comparison with Anstreicher and Puges [2].

A.3 Closedness of the PSD–nonnegative decomposition

The order-four cone identity used in Theorem 4.3 is an equality of cones, rather than a closure statement. The following elementary closedness argument also explains why dualizing $\text{CP}_4 = \text{DNN}_4$ does not introduce a missing limit case.

Lemma A.3. *For every r , the cone $\mathbb{S}_+^r + \mathcal{N}_r$ is closed, where $\mathcal{N}_r$ is the symmetric entrywise nonnegative cone.*

Proof. Suppose $P_j + N_j \rightarrow A$, with $P_j \succeq 0$ and $N_j \geq 0$ entrywise. Their diagonal entries are nonnegative and bounded, since their sums converge. The PSD inequality $|(P_j)_{ab}|^2 \leq (P_j)_{aa}(P_j)_{bb}$ bounds every entry of P_j . Then $N_j = (P_j + N_j) - P_j$ is bounded as well. A common convergent subsequence yields $A = P + N$, with $P \succeq 0$ and $N \geq 0$ entrywise. $\square$

B Coefficient reconstruction for boundary rays

This appendix proves the rank statement used in Theorem 5.11 by coefficient elimination. No generic-position assumption or computational rank test is needed.

The family has the following exact edge restrictions, where $D = d_1 + d_2 - h$:

$$\begin{aligned} q(x, 0, 0) &= (h - d_1 x)^2, & q(0, y, 0) &= (h - d_2 y)^2, \\ q(1, 0, z) &= (d_3 z - (d_1 - h))^2, & q(0, 1, z) &= (d_3 z - (d_2 - h))^2, \\ q(1, 1, z) &= (d_3 z - (D + k))^2. \end{aligned}$$

These identities verify every edge value and tangency used below. At $h = 0$, the origin additionally has zero x - and y -derivatives. At $h = d_1$, the vertex $(1, 0, 0)$ has zero x - and z -derivatives.

Write a candidate quadratic as

$$p = A + ax + by + cz + ux^2 + vy^2 + wz^2 + rxy + sxz + tyz. \quad (49)$$

Let

$$\gamma = \frac{d_1 - h}{d_3}, \quad \delta = \frac{d_2 - h}{d_3}, \quad \eta = \frac{d_1 + d_2 - h + k}{d_3}.$$

In every regime under consideration, the three z -edges at $(1, 0)$, $(0, 1)$, and $(1, 1)$ carry a double zero at γ , δ , and η , respectively. The term “double zero” means that both the value and the edge derivative vanish; it includes an endpoint with the stated tangency. The corresponding restrictions are therefore

$$p(1, 0, z) = w(z - \gamma)^2, \quad (50)$$

$$p(0, 1, z) = w(z - \delta)^2, \quad (51)$$

$$p(1, 1, z) = w(z - \eta)^2. \quad (52)$$

These are identities even if $w = 0$. Their linear coefficients give

$$c = 2w(\eta - \gamma - \delta) = \frac{2w(h + k)}{d_3}, \quad s = -\frac{2w(d_1 + k)}{d_3}, \quad t = -\frac{2w(d_2 + k)}{d_3}. \quad (53)$$

Their constant coefficients give

$$A + a + u = w\gamma^2, \quad A + b + v = w\delta^2, \quad A + a + b + u + v + r = w\eta^2. \quad (54)$$

B.1 The upper contact reaches the vertex

Suppose $0 < h < \min(d_1, d_2)$ and $d_3 = d_1 + d_2 - h + k$, so $\eta = 1$. The additional x - and y -edge contacts at h/d_1 and h/d_2 are interior double zeros. Thus

$$A = u \frac{h^2}{d_1^2} = v \frac{h^2}{d_2^2}, \quad a = -\frac{2uh}{d_1}, \quad b = -\frac{2vh}{d_2}.$$

The first equation in (54) becomes

$$\frac{A(d_1 - h)^2}{h^2} = \frac{w(d_1 - h)^2}{d_3^2}.$$

Since $d_1 > h$, cancellation is valid. Hence

$$A = \frac{wh^2}{d_3^2}, \quad u = \frac{wd_1^2}{d_3^2}, \quad v = \frac{wd_2^2}{d_3^2}, \quad a = -\frac{2whd_1}{d_3^2}, \quad b = -\frac{2whd_2}{d_3^2}.$$

Together with (53), the final constant equation determines r . All ten coefficients are fixed by w .

B.2 The two lower contacts reach the origin

Suppose $h = 0$ and $d_3 \geq d_1 + d_2 + k$. At the origin, the value and the x - and y -derivatives are zero, so $A = a = b = 0$. The first two equations in (54) give

$$u = \frac{wd_1^2}{d_3^2}, \quad v = \frac{wd_2^2}{d_3^2}.$$

Equations (53) and (54) determine the remaining coefficients. There is no division by d_1 or d_2 , so this proof includes either or both of them equal to zero. In that event, a zero on a z -edge reaches its lower vertex and its z -derivative remains zero. If $d_3 = d_1 + d_2 + k$, the upper contact reaches $(1, 1, 1)$ with its z -derivative zero. Thus the same equations apply at both ends of this parameter regime.

B.3 Two contacts merge at a lower vertex

Suppose $h = d_1 < d_2$ and $d_3 \geq d_2 + k$. The contact at $(1, 0, 0)$ has zero value and zero x - and z -derivatives. Consequently,

$$A = u, \quad a = -2u, \quad c + s = 0.$$

The y -edge contact is at $\beta = d_1/d_2 < 1$ and has zero derivative. For $d_1 = 0$, this is the origin with a y -tangency. In both cases,

$$A = v\beta^2, \quad b = -2v\beta.$$

The second equation of (54) becomes

$$v(1 - \beta)^2 = \frac{w(d_2 - d_1)^2}{d_3^2}.$$

Since $d_2 > d_1$, it follows that

$$v = \frac{wd_2^2}{d_3^2}, \quad A = u = \frac{wd_1^2}{d_3^2}, \quad a = -\frac{2wd_1^2}{d_3^2}, \quad b = -\frac{2wd_1d_2}{d_3^2}.$$

The formulas for c, s, t and r follow as before. This argument also covers $d_1 = 0$, because it divides only by $d_2 - d_1$ and $d_2 > 0$. The upper contact can be either an interior point of the z -edge or its upper vertex, depending on whether $d_3 > d_2 + k$ or equality holds.

B.4 The common one-dimensional kernel

In all three regimes, the resulting vector of coefficients is

$$(A, a, b, c, u, v, w, r, s, t) = \frac{w}{d_3^2} (h^2, -2hd_1, -2hd_2, 2d_3(h+k), d_1^2, d_2^2, d_3^2, \\ 2d_1d_2 + k(2(d_1 + d_2 - h) + k), -2d_3(d_1 + k), -2d_3(d_2 + k)).$$

This is exactly $(w/d_3^2)q_{h,d,k}$. Since $d_3 > 0$ in every stated regime, the family member is nonzero and supplies a kernel vector. Every kernel vector has the displayed form, including the zero vector when $w = 0$. The contact system therefore has rank nine in the ten-dimensional quadratic space throughout each regime. This proves the claim used in Theorem 5.11.

C Details of the semidefinite representability arguments

C.1 The real closed field inputs and their use

We record the external results needed in Theorems 6.2 and 6.3. This also distinguishes the copositive nonrepresentability theorem from the additional transfer to the box.

A field is *real closed* if it is ordered and every positive element has a square root and every polynomial of odd degree has a root. The transfer principle for real closed fields says that a first-order formula with real parameters holds over $\mathbb{R}$ if and only if it holds over any real closed extension of $\mathbb{R}$. In particular, copositivity, box nonnegativity of a quadratic, and the equivalence of a semialgebraic set with a fixed finite semidefinite lift transfer to such an extension. Positive semidefiniteness can be expressed by the nonnegativity of finitely many principal minors.

For a real closed field F and a divisible ordered abelian group Γ , the Hahn field $F((\varepsilon^\Gamma))$ consists of formal series $\sum_{a \in \Gamma} c_a \varepsilon^a$ with well-ordered support. The sign of a nonzero series is the sign of its coefficient at the least exponent. This field is real closed [3, Section 2]. Its group algebra $F[\Gamma]$ consists of the finite such sums, regarded without reference to the ordering.

Call a function $f : \Gamma \rightarrow F$ *positive definite* here if, for every list of distinct $a_1, \dots, a_r \in \Gamma$, the matrix $(f(a_i + a_j))_{i,j}$ is positive definite. The use of sums of indices, rather than differences, follows the ordered-field setting of Bodirsky et al. [3].

Lemma C.1 (External inputs from Bodirsky–Kummer–Thom). *The following statements hold.*

- (i) *For the Horn form $h(v) = v^\top H v$, the polynomial $p(z) = h(z_1^2, \dots, z_5^2)$ is nonnegative on $\mathbb{R}^5$, but $p(z_1^d, \dots, z_5^d)$ is not a sum of squares for any integer $d \geq 1$. Consequently p is not a sum of squares in $\mathbb{R}[\mathbb{Q}^5]$.*
- (ii) *There are a real closed extension $F \supseteq \mathbb{R}$ and an F -linear functional $T : F[\mathbb{Q}^5] \rightarrow F$ such that $T(a^2) > 0$ for every nonzero $a \in F[\mathbb{Q}^5]$ and $T(p) < 0$.*
- (iii) *If Γ is divisible and ordered, F is real closed, and $f : \Gamma \rightarrow F$ is positive definite, the map $L_f(\sum_a c_a \varepsilon^a) = \sum_a f(a) c_a \varepsilon^a$ on $F((\varepsilon^\Gamma))$ is F -linear and completely positive: entrywise application preserves positive semidefinite matrices of every size. If $f(0) = 1$, this map is unital.*

Statement (i) uses Example 3.4 and Remark 3.2, statement (ii) is the positive-square separation in Theorem 3.7 as applied in Proposition 3.11, and statement (iii) is Theorem 2.13(1) of Bodirsky et al. [3].

Only these external inputs, transfer for real closed fields, and closed-cone duality are used in the cube obstruction. Here is the passage from them to (38). Normalize T by its positive value $T(1)$ and set

$$f(a) = \frac{T(z^a)}{T(1)}, \quad a \in \mathbb{Q}^5.$$

Then $f(0) = 1$. For distinct a_i and a nonzero vector $(c_i) \in F^r$,

$$\sum_{i,j} c_i c_j f(a_i + a_j) = \frac{T((\sum_i c_i z^{a_i})^2)}{T(1)} > 0,$$

so f is positive definite. Since $p = \sum_{i,j} H_{ij} z^{2e_i + 2e_j}$, its negative value under T gives (38). The order on $\mathbb{Q}^5$ does not affect this construction of f , so it may be chosen lexicographically afterward. The coefficient map of part (iii) is then the map used in (39).

For clarity, the preservation of a semidefinite lift under this map has a short direct proof. Suppose a semialgebraic set $S \subseteq \mathbb{R}^d$ has the lift (4), and let E be a real closed extension. Transfer says that the same existential matrix formula describes $S(E)$. If (y, z) is a feasible witness over E , a unital real-linear completely positive map $L : E \rightarrow E$ gives

$$0 \preceq L \left(A_0 + \sum_i y_i A_i + \sum_j z_j B_j \right) = A_0 + \sum_i L(y_i) A_i + \sum_j L(z_j) B_j.$$

Thus the coordinatewise image $L(y)$ belongs to $S(E)$. Unitality fixes the constant matrix, real-linearity fixes its real coefficients, and complete positivity preserves the matrix inequality. No order preservation of multivariate polynomial evaluation is being assumed: the negative evaluation in (41) is computed after mapping the coefficients.

We also give an elementary verification that H is copositive. For $v \geq 0$,

$$v^\top H v = \left(\sum_i v_i \right)^2 - 4 \sum_{ij \in E(C_5)} v_i v_j.$$

Fix the sum of the coordinates and maximize the edge sum on this compact simplex. Among its maximizers choose one with the smallest support. If two nonadjacent vertices have positive weight, move their combined weight to the one with the larger weighted neighbor sum. Their mutual product does not occur, so this does not reduce the edge sum, and it reduces the support. This is a contradiction. The support of a minimal maximizer is therefore a clique, which has at most two vertices in C_5 . For such a support the edge sum is at most one quarter of the squared coordinate sum. Hence $v^\top H v \geq 0$. This verifies the copositivity input without any numerical test.

C.2 The simple-ray geometric condition

Let $K \subseteq \mathbb{R}^m$ be full dimensional and pointed, and let r span a simple extreme ray. Write $\ell_2, \dots, \ell_m$ for inner facet forms of the $m - 1$ incident facets. They are linearly independent: their common kernel is the linear span of the ray, since intersecting the incident facets recovers the ray. Choose ℓ_1 strictly positive on $K \setminus \{0\}$. Such a form exists for a pointed full-dimensional polyhedral cone. It is not in the span of the incident facet forms because it does not vanish on r . Thus

$$A : z \mapsto (\ell_1(z), \dots, \ell_m(z))$$

is invertible. Normalize r so that $\ell_1(r) = 1$, and put $v_k = A^{-1}e_k$ for $k = 2, \dots, m$.

Every nonincident facet form g_j satisfies $g_j(r) > 0$. Choose $\delta > 0$ such that, for every such facet,

$$\delta \sum_{k=2}^5 |g_j(v_k)| \leq \frac{1}{2} g_j(r). \quad (55)$$

If there are no nonincident facets, any positive δ works. For a vector $u \in \mathbb{R}_+^5$ with $u_k \leq \delta u_1$, first suppose $u_1 > 0$ and set

$$z = u_1 r + \sum_{k=2}^5 u_k v_k.$$

Its incident facet values are $u_2, \dots, u_5, 0, \dots, 0$, all nonnegative. For a nonincident facet, (55) gives

$$g_j(z) \geq u_1 g_j(r) - \delta u_1 \sum_{k=2}^5 |g_j(v_k)| \geq \frac{1}{2} u_1 g_j(r) > 0.$$

Thus $z \in K$, and $(\ell_1, \dots, \ell_5)(z) = u$. When $u_1 = 0$, all of u is zero and $z = 0$ suffices. Each ℓ_i is nonnegative on K , so this proves both inclusions in (42).

For the polytope consequence, use intrinsic affine coordinates if the polytope is not full dimensional in its given ambient space. The cone $K = \text{cone}(\{1\} \times X)$ is closed and pointed, with dimension $\dim X + 1$. Its facets incident to the ray through $(1, v)$ correspond exactly to facets of X incident to v . The map from symmetric forms on K to quadratic polynomials on X , obtained by setting the first coordinate to 1, is a linear isomorphism. This proves the correspondence used in Corollary 6.4.

C.3 The zero sets that describe proper faces

Let $p \geq 0$ be a nonzero quadratic on C_4 , and let F be a face of the cube whose relative interior contains a zero a of p . In affine coordinates on F , local nonnegativity at a implies zero gradient and a positive semidefinite quadratic part A_F . Since p is quadratic,

$$p(a + v) = v^\top A_F v$$

holds throughout the affine hull of F . Its zero set on F is therefore

$$Z_F = F \cap (a + \ker A_F),$$

a polytope. Every cube zero lies in the relative interior of a unique cube face, so the whole zero set is the union of the finitely many Z_F obtained in this manner. Each has dimension at most three. For boundary faces this follows from their dimension. For the full cube, A_F cannot be zero, because zero gradient and zero value at a would then imply $p = 0$.

To verify that the associated face of $\mathcal{Q}_4$ is precisely the convex hull of the moments over this zero set, use a finite convex representation by cube atoms. A nonnegative sum of their p -values is zero exactly when every positive-weight atom has p -value zero. Thus

$$\{(m, Y) \in \mathcal{Q}_4 : \langle p, (1, m, Y) \rangle = 0\} = \text{conv}\{(x, xx^\top) : x \in \bigcup_F Z_F\}.$$

For a polytope of affine dimension at most three, triangulation and $\text{CP}_r = \text{DNN}_r$ for $r \leq 4$ give a finite moment lift. An affine change of coordinates $x = a + Bz$ acts linearly on the homogenized moments $(1, z, zz^\top)$, so this conclusion applies to each Z_F in its original coordinates.

If compact sets $S_1, \dots, S_r$ have finite lifts, Lemma 6.1 with $D = \{0\}$ gives finite lifts of their closed homogenized cones. The system

$$(t_i, y_i) \in \widehat{S_i}, \quad \sum_i t_i = 1, \quad y = \sum_i y_i$$

is a finite lift of $\text{conv}(\bigcup_i S_i)$. At zero weight its visible contribution is zero; at positive weight it contributes a weighted point of S_i . This justifies the convex-hull step even when the original lifts have unbounded auxiliary variables.

Finally, let F be an arbitrary proper face of a compact convex set S in its affine hull, and take $a \in \text{relint } F$. A proper supporting hyperplane of S through a cuts out an exposed proper face E containing F . The inclusion follows because a linear function attaining its minimum at a relative interior point of F is constant on F . If $F \neq E$, it is a proper face of E , so repeat the argument in the affine hull of E . At every step the containing face decreases in dimension. Eventually it equals F . After the first step, all these affine sections preserve the finite lift already established for E . This proves Proposition 6.5 for nonexposed faces as well.

D Separation and comparison tools

This appendix gives the exact statements behind the numerical comparisons. The statements concern exact arithmetic. The archived implementations use floating point, so their reported values need the qualifications in Section 8.5.

D.1 Family separation on the simplex

Write $\Delta_S = \{v \in \mathbb{R}_+^S : \mathbf{1}^T v = 1\}$ for a nonempty $S \subseteq \{1, 2, 3, 4\}$. For $A \in \mathbb{S}^4$, define

$$\mu(A) = \min_{v \in \Delta_{\{1,2,3,4\}}} v^T A v, \quad K_S = \begin{pmatrix} A_{SS} & \mathbf{1} \\ \mathbf{1}^T & 0 \end{pmatrix}. \quad (56)$$

The normalization in (56) differs from the Euclidean normalization in Proposition 4.4; both detect exactly whether A is copositive.

Proposition D.1 (Fifteen-support separation). *For each nonempty support S , solve*

$$K_S \begin{pmatrix} v_S \\ \nu \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (57)$$

Retain a solution whenever $v_S \geq 0$, and extend it by zero outside S . The least objective value among these feasible stationary solutions is $\mu(A)$. If a system is singular but has feasible solutions, all of them have the same objective value. Moreover, a global minimizer occurs on a support for which K_S is nonsingular. Thus singular systems may be omitted in exact arithmetic without losing the minimum.

Proof. Every retained vector is feasible for (56), so its objective is at least $\mu(A)$. A global minimizer exists by compactness. Choose one, v^* , with the smallest support S . It lies in the relative interior of Δ_S , so first-order stationarity on its affine hull gives $A_{SS}v_S^* = \lambda \mathbf{1}$. Hence $(v_S^*, -\lambda)$ solves (57).

Suppose K_S were singular. There would be a nonzero pair (d, η) with $A_{SS}d + \eta \mathbf{1} = 0$ and $\mathbf{1}^T d = 0$. Necessarily $d \neq 0$. For every $t \in \mathbb{R}$,

$$(v_S^* + td)^T A_{SS}(v_S^* + td) = (v_S^*)^T A_{SS}v_S^* + 2t\lambda \mathbf{1}^T d - t^2 \eta \mathbf{1}^T d = \mu(A).$$

Because d has both positive and negative coordinates and $v_S^* > 0$, one can move in this direction until a coordinate first becomes zero. The resulting point is feasible and has a smaller support, a contradiction. Thus K_S is nonsingular and v^* is among the retained candidates.

Finally, if $u, v \in \Delta_S$ solve the stationary equations with $A_{SS}u = \lambda \mathbf{1}$ and $A_{SS}v = \rho \mathbf{1}$, symmetry gives $\lambda = v^T A_{SS}u = u^T A_{SS}v = \rho$. Their objective values are therefore equal. This also handles degenerate faces with infinitely many stationary solutions. No positive-definiteness assumption on A_{SS} is needed: nonminimizing stationary points remain feasible candidates and cannot lower the minimum below $\mu(A)$. $\square$

For the family of Section 4, take $A = B - bb^T$ from (4.3). Minimization over the unrestricted parameter h gives $h = -b^T v$ and family value $v^T A v$. A negative value in (56) therefore gives a violated family inequality, with $(d_1, d_2, d_3, k) = v$. Any vector $v \geq 0$ gives a valid inequality, even if it is not a minimizer. The archived implementation clips tiny negative coordinates, renormalizes the vector, and evaluates the resulting feasible candidate. This preserves the mathematical validity of its parameter choice, but skipping a nearly singular system can miss a violation. A floating-point value of $\mu(A)$ is consequently a separation diagnostic, not a certificate of copositivity.

D.2 The five-tetrahedron lift and normalized hull depth

For $u \in C_3$, put $\bar{u} = (1, u)^T$, and write

$$\mathcal{M}_3 = \text{conv}\{\bar{u}\bar{u}^T : u \in C_3\}.$$

This is the matrix form of $\mathcal{Q}_3$. Let

$$M_c = \begin{pmatrix} 1 & 1/2 & 1/2 & 1/2 \\ 1/2 & 1/3 & 1/4 & 1/4 \\ 1/2 & 1/4 & 1/3 & 1/4 \\ 1/2 & 1/4 & 1/4 & 1/3 \end{pmatrix}, \quad (58)$$

the moment matrix of the uniform distribution on C_3 . Use the following five tetrahedra; strings denote cube vertices:

$$\begin{aligned} (000, 110, 101, 011), & \quad (100, 000, 110, 101), \\ (010, 000, 110, 011), & \quad (001, 000, 101, 011), \\ (111, 110, 101, 011). & \end{aligned} \quad (59)$$

Let $\bar{A}_p$ have the four homogenized vertices of tetrahedron p as columns. To see that these tetrahedra cover the cube, the central tetrahedron has barycentric coordinates

$$1 - \frac{x+y+z}{2}, \quad \frac{x+y-z}{2}, \quad \frac{x+z-y}{2}, \quad \frac{y+z-x}{2}.$$

Outside it, one of $x > y + z$, $y > x + z$, $z > x + y$, or $x + y + z > 2$ holds. These regions are the four remaining tetrahedra, and their interiors are disjoint.

The exact lift of Anstreicher and Burer [1] specializes to

$$\mathcal{M}_3 = \left\{ \sum_{p=1}^5 \bar{A}_p W_p \bar{A}_p^T : W_p \in \text{DNN}_4, \quad \sum_{p=1}^5 \mathbf{1}^T W_p \mathbf{1} = 1 \right\}. \quad (60)$$

For completeness, $\text{DNN}_4 = \text{CP}_4$: factor each W_p into a sum of ww^T with $w \geq 0$. For $w \neq 0$, let $\alpha = \mathbf{1}^T w$ and u be the point of tetrahedron p with $\bar{u} = \bar{A}_p w / \alpha$. Then $\bar{A}_p w w^T \bar{A}_p^T = \alpha^2 \bar{u} \bar{u}^T$, and the normalization in (60) says that these weights sum to one. Conversely, every cube point belongs to a tetrahedron and has nonnegative barycentric coordinates, yielding a feasible rank-one W_p .

Proposition D.2 (Hull depth and exact cut correction). *For $M \in \mathbb{S}^4$ with $M_{00} = 1$, define*

$$\begin{aligned} \delta(M) = \min \quad & \langle C, M \rangle \\ \text{subject to} \quad & \langle C, M_c \rangle = 1, \\ & \bar{A}_p^T C \bar{A}_p - N_p \succeq 0 \quad (p = 1, \dots, 5), \\ & N_p \geq 0 \text{ entrywise}, \quad \text{diag}(N_p) = 0. \end{aligned} \quad (61)$$

The minimum is finite and attained. The projected feasible matrices C are exactly the quadratics $q_C(u) = \bar{u}^T C \bar{u}$ that are nonnegative on C_3 and have uniform mean one. Consequently,

$$\delta(M) < 0 \iff M \notin \mathcal{M}_3.$$

If $\varepsilon \geq \max\{0, -\delta(M)\}$, then

$$\frac{M + \varepsilon M_c}{1 + \varepsilon} \in \mathcal{M}_3. \quad (62)$$

For any symmetric C , the shifted quadratic $q_C + s$ is nonnegative on C_3 whenever

$$s \geq \max \left\{ 0, -\min_{1 \leq p \leq 5} \mu(\bar{A}_p^T C \bar{A}_p) \right\}. \quad (63)$$

Proof. On tetrahedron p , $q_C(u) = \lambda^T \bar{A}_p^T C \bar{A}_p \lambda$ for $\lambda \geq 0$ with $\mathbf{1}^T \lambda = 1$. Thus q_C is nonnegative on the cube exactly when all five matrices $\bar{A}_p^T C \bar{A}_p$ are copositive. The equality $\text{COP}_4 = \mathbb{S}_+^4 + (\mathbb{R}_+^{4 \times 4} \cap \mathbb{S}^4)$ gives precisely the constraints in (61); any nonnegative diagonal component can be moved into the positive semidefinite component.

The cone of cube-nonnegative quadratics is closed. A nonzero element has strictly positive uniform mean: a continuous nonnegative polynomial with zero integral vanishes on the cube and hence is the zero polynomial. Its mean-one slice is compact. Indeed, an unbounded sequence in this slice, divided by its coefficient norm, would have a subsequence converging to a nonzero nonnegative quadratic of mean zero, a contradiction. The slice is nonempty because it contains the constant one. Therefore (61) has a finite attained minimum in C , and a copositive decomposition supplies the auxiliary N_p .

Every $M \in \mathcal{M}_3$ evaluates all nonnegative quadratics nonnegatively. If M is outside this compact convex hull, strict separation gives a matrix D and a scalar a with $\langle D, M \rangle < a \leq \langle D, \bar{u}\bar{u}^T \rangle$ for every cube point. Subtracting a from the constant term yields a nonzero cube-nonnegative quadratic with negative value at M , because $M_{00} = 1$. Its positive uniform mean permits normalization in (61). This proves the equivalence.

For every normalized nonnegative quadratic, $\langle C, M + \varepsilon M_c \rangle \geq \delta(M) + \varepsilon \geq 0$. The preceding separation characterization and the leading entry $1 + \varepsilon$ give (62). Finally, on tetrahedron p , $q_C(u) \geq \mu(\bar{A}_p^T C \bar{A}_p)$, proving (63). $\square$

Formula (63) is an exact certification rule if its five simplex minima are evaluated exactly, or if certified lower bounds replace them. Applying the floating-point routine of Proposition D.1 with a small rounding margin does not establish this rule in exact arithmetic.

D.3 An upper bound on triple-local improvement

Let y be a vector of first and second moments on a fixed pattern, with every diagonal recorded. For a recorded triple T , write $M_T(y)$ for its normalized 4×4 moment matrix. Let y_c be the restriction of the uniform cube moments to this pattern. Thus each $M_T(y_c)$ equals (58).

Proposition D.3 (Triple-local gain bound). *Let R be a convex set of moment vectors containing y_c , let $y^* \in R$, and let f be an affine objective. Let $\mathcal{T}$ be a set of recorded triples and assume*

$$\varepsilon \geq \max_{T \in \mathcal{T}} \max\{0, -\delta(M_T(y^*))\}.$$

For each T , add any system involving its moments and auxiliaries used only by that triple, whose projection contains $\mathcal{M}_3$. Let R' be the resulting moment feasible set. Then

$$\inf_{y \in R'} f(y) \leq \frac{f(y^*) + \varepsilon f(y_c)}{1 + \varepsilon} = f(y^*) + \frac{\varepsilon}{1 + \varepsilon} (f(y_c) - f(y^*)). \quad (64)$$

If y^ minimizes f over R , the last term bounds the improvement of the relaxation value.*

Proof. Convexity gives $y_\varepsilon = (y^* + \varepsilon y_c)/(1 + \varepsilon) \in R$. By Proposition D.2, $M_T(y_\varepsilon)$ belongs to $\mathcal{M}_3$ for every $T \in \mathcal{T}$. Each added system has feasible auxiliaries at these moments. Because the auxiliaries are independent between triples, they can all be chosen at once. Hence $y_\varepsilon \in R'$. Evaluating f there proves (64). $\square$

This argument applies unchanged to clique moment patterns. It covers the family blocks, tetrahedral lifts, and the Anstreicher–Puges systems with their own trilinear moment for each triple. It also covers the disjoint-support systems when their additional moments are not identified across triples. It does not cover a common auxiliary such as $x_i^2 x_j$ used by two different triples: local representing measures need not give the same value to that shared auxiliary.

For the objective $f(m, Y) = \langle H, Y \rangle + g^T m$,

$$f(y_c) = \frac{1}{3} \sum_i H_{ii} + \frac{1}{4} \sum_{i \neq j} H_{ij} + \frac{1}{2} \sum_i g_i.$$

An application to a numerical y^* requires actual feasibility for R and a one-sided upper bound on every $-\delta(M_T(y^*))$. If L_R is a reported lower bound on the baseline value, the comparison with L_R must also include the margin $f(y^*) - L_R$.

D.4 Dual residual correction

Proposition D.4. *Consider $\inf\{c^T u : Au + s = b, s \in K\}$, where every feasible u satisfies $\ell_i \leq u_i \leq h_i$. For any $z \in K^*$, put $r = c + A^T z$. Then every feasible point satisfies*

$$c^T u \geq -b^T z + \sum_i \min\{r_i \ell_i, r_i h_i\}. \quad (65)$$

Proof. Since $\langle z, s \rangle \geq 0$, $c^T u = r^T u - z^T A u = r^T u - b^T z + \langle z, s \rangle \geq r^T u - b^T z$. Minimize each component of $r^T u$ over its interval. $\square$

The implementation projects the approximate dual vector onto the relevant cones and subtracts rounding allowances before using (65). The variables have explicit finite bounds: recorded moments lie in $[0, 1]$, the six family auxiliaries lie in $[0, 3]$, and tetrahedral weights lie in $[0, 1]$. The family bound is implied by the block at a baseline-feasible point: each entry of B and b has absolute value at most one, while $S = B - N - b b^T \succeq 0$ has diagonal at most one and hence entries of absolute value at most one. Thus $N_{ij} = B_{ij} - b_i b_j - S_{ij} \leq 3$. For the tetrahedral lift, nonnegativity and $\sum_p \mathbf{1}^T W_p \mathbf{1} = 1$ imply each entry of W_p is at most one. These restrictions preserve the intended lifts. The reported corrected values are careful floating-point estimates. They become rigorously certified lower bounds only if cone membership, residuals, and the rounding allowances are themselves verified with rigorous arithmetic.

E Proof of the edge-contact classification

Throughout this appendix, $q(x) = q_0 + a^\top x + x^\top Qx$ satisfies the hypotheses of Theorem 5.12, and $d_i = \sqrt{Q_{ii}} > 0$. We assume that q is not an affine square. Coordinate complementation preserves vertex positivity, square coefficients, and extremality. We first exclude a nonpositive mixed-coefficient product, then make every mixed coefficient positive.

E.1 The nonpositive-product sign class

If $c_{12}c_{13}c_{23} \leq 0$, coordinate complementation makes all mixed coefficients nonpositive, as in the proof of Theorem 5.6. The exactness theorem of Burer et al. [6, Theorem 1] applies to the SDP

$$\min\{\mathcal{L}_{m,Y}(q) : \begin{pmatrix} 1 & m^\top \\ m & Y \end{pmatrix} \succeq 0, Y_{ij} \leq m_i \ (i, j = 1, 2, 3)\}.$$

Its feasible set is compact. Indeed, the moment-matrix constraint gives $Y_{ii} \geq m_i^2$, and $Y_{ii} \leq m_i$ gives $0 \leq m_i \leq 1$. The principal 2×2 moment blocks bound every $|Y_{ij}| \leq \sqrt{Y_{ii}Y_{jj}} \leq 1$. The uniform cube moments $m_i = 1/2$, $Y_{ii} = 1/3$, $Y_{ij} = 1/4$ for $i \neq j$ satisfy all inequalities strictly and have a positive-definite moment matrix. Slater's condition therefore gives an attained dual with the same value.

Put $\bar{x} = (1, x^\top)^\top$. Matching coefficients in the SDP Lagrangian gives the dual

$$\max \left\{ \beta : q(x) = \beta + \bar{x}^\top Z \bar{x} + \sum_{i,j=1}^3 \lambda_{ij} x_i(1 - x_j), \quad Z \succeq 0, \lambda_{ij} \geq 0 \right\}.$$

The displayed identity is a polynomial identity. Its optimum is $\beta = \min_{C_3} q \geq 0$ by the exactness theorem. Thus q is a sum of affine squares, a nonnegative constant, and nonnegative multiples of $x_i(1 - x_j)$, including the caps when $i = j$.

Extremality makes every nonzero summand proportional to q . Every product $x_i(1 - x_j)$ vanishes at a vertex, whereas q is positive at all vertices, so its multiplier must be zero. A nonzero constant cannot be proportional to q because q has positive square coefficients. The PSD term therefore makes q an affine square, contrary to our assumption. Consequently the mixed-coefficient product is positive, and a coordinate complementation makes all three mixed coefficients positive for the rest of the proof.

E.2 Zeros and perturbations

A zero in the cube interior requires the full Hessian to be positive semidefinite and the gradient to vanish. The quadratic is then a sum of affine squares; extremality would make it a single affine square. A zero in a facet interior is stationary in both facet coordinates and requires the facet block to be positive semidefinite. This is excluded by Lemma F.2, proved in Appendix F. Vertex zeros are excluded by hypothesis. Thus every zero is in an edge interior, and there is at most one on each edge because its quadratic coefficient is positive.

At a zero in an edge parallel to coordinate i , the derivative in that coordinate vanishes. The two derivatives directed into adjacent facets are nonnegative. They are in fact positive. If the inward derivative in coordinate j were zero, that zero would be stationary in both coordinates of the adjacent facet. Its facet block would have to be positive semidefinite: any negative quadratic direction can be reversed to point inward in coordinate j , while coordinate i admits both signs near the edge-interior zero. Lemma F.2 excludes this stationary facet zero for a non-square extreme ray.

Let m be the number of edge zeros. Impose on a quadratic perturbation p its value and edge derivative being zero at each zero of q . These are $2m$ homogeneous linear equations. Every such p satisfies $q \pm \epsilon p \geq 0$ for sufficiently small $\epsilon > 0$. To see this locally, use an edge displacement u and inward displacements $s, t \geq 0$. The positive edge curvature and strict inward derivatives give, in a sufficiently small neighborhood,

$$q \geq \delta(u^2 + s + t), \quad |p| \leq M(u^2 + s + t)$$

for some $\delta, M > 0$. The second bound uses the zero value and edge derivative of p ; its other linear terms are bounded by multiples of $s + t$, and its remaining quadratic terms by multiples of $u^2 + s + t$. The finitely many neighborhoods give a common permissible ϵ . On their compact complement, q has a positive minimum, so reducing ϵ also controls p there.

If $m \leq 4$, the solution space of the perturbation equations has dimension at least $10 - 2m \geq 2$. It contains q and a nonproportional p , contradicting extremality. Consequently $m \geq 5$. This argument uses an upper bound on the number of independent equations; it does not assume that a contact system has full rank.

E.3 Two local restrictions on the contact graph

For a vertex v , put $f_v = \sqrt{q(v)} > 0$. If the edge vw is parallel to coordinate i and has an interior zero, its restriction is a square and

$$f_v + f_w = d_i. \tag{66}$$

Indeed its zero divides the unit edge into lengths f_v/d_i and f_w/d_i . Conversely, (66) with both endpoint values positive implies an interior zero: the endpoint values and the quadratic coefficient determine that square restriction.

Suppose two contact edges at v use coordinates i, j . Reflect their coordinates so that both point positively from v . Writing their zero positions as $\alpha_i = f_v/d_i$ and $\alpha_j = f_v/d_j$, the restriction to their face has the form

$$f_v^2 \left(1 - \frac{u_i}{\alpha_i} - \frac{u_j}{\alpha_j} \right)^2 + (\hat{c}_{ij} - 2d_i d_j) u_i u_j,$$

where $\hat{c}_{ij} = (-1)^{v_i + v_j} c_{ij}$. At $(\alpha_i/2, \alpha_j/2)$, nonnegativity implies $\hat{c}_{ij} \geq 2d_i d_j$. Since $c_{ij} > 0$, this forces $v_i = v_j$. The inequality need not be strict at this stage.

Three contact edges at one vertex are also impossible for an extreme q . The same calculation in three coordinates gives an identity

$$q = f_v^2 \left(1 - \sum_i \frac{u_i}{\alpha_i} \right)^2 + \sum_{i < j} (\hat{c}_{ij} - 2d_i d_j) u_i u_j.$$

Every summand is nonnegative on the reflected cube. If every product coefficient is zero, q is the excluded affine square. Otherwise this is a nontrivial decomposition of q into nonnegative quadratics, contrary to extremality.

We shall also use a two-variable observation. Suppose a nonnegative quadratic on a square has positive mixed coefficient and has interior zeros on two opposite edges parallel to coordinate i . Let their zero positions be $\alpha, \beta \in (0, 1)$, at coordinate $j = 0, 1$, respectively. Then

$$q(u_i, u_j) = d_i^2 (u_i - \alpha - (\beta - \alpha)u_j)^2 + \eta u_j (1 - u_j), \quad \eta = d_i^2 (\beta - \alpha)^2 - d_j^2.$$

The mixed coefficient being positive gives $\alpha > \beta$. Evaluating at the affine minimizer in u_i , which stays in $(0, 1)$, gives $\eta \geq 0$. Thus

$$d_j \leq d_i(\alpha - \beta) < d_i. \tag{67}$$

E.4 The possible contact graphs

Let the Hamming weight of a vertex be its number of unit coordinates. The rule $v_i = v_j$ shows that, when two contact edges meet, their other endpoints have equal Hamming weight. Hence a connected component of the contact graph stays between one pair of adjacent weight levels. The cube edges between levels 0, 1 form a three-edge star, those between levels 1, 2 form a six-edge cycle, and those between levels 2, 3 form a second three-edge star. Every vertex has contact degree at most two.

Write a, b for the numbers of contact edges in the lower and upper stars, and m_0 for the number in the central cycle. Thus $0 \leq a, b \leq 2$ and $a + b + m_0 \geq 5$. A leaf of an outer star cannot also meet a central contact edge, because that would join different pairs of weight levels. Up to coordinate permutation and simultaneous complementation of all three coordinates, the following cases exhaust the possibilities.

- (i) $a = b = 0$. There are at least five central-cycle edges.
- (ii) $a = 1, b = 0$. Deleting the level-one endpoint of the lower edge leaves a four-edge path in the central cycle. All four edges must be present.
- (iii) $a = b = 1$. Deleting their central endpoints leaves at most a three-edge path. To reach five contacts, the endpoints must be adjacent in the cycle and all three remaining edges must be present.
- (iv) $a = 2, b = 1$. Only one level-one vertex remains available centrally, so the central contacts must be its two incident edges. The upper-star endpoint is its nonneighbor in level two.
- (v) $a = b = 2$. Only one vertex in each central level remains available. They must be adjacent, and their edge must be present.

The case $a = 2, b = 0$ cannot reach five contacts, because at most two central edges remain available. Interchanging the outer stars accounts for the cases with $b > a$ not listed above. These observations also cover graphs with more than five contacts: the central-only case allows six, and the availability bounds in every other case allow at most five.

E.5 Excluding the central cycle and three other cases

In case (i), five equations (66) force the sixth. To verify this, list the six directions around the central cycle as 1, 3, 2, 1, 3, 2. The alternating sum of the six left sides is zero, and so is the alternating sum of their right sides. Both endpoint square roots are positive, so the missing edge also has an interior zero.

Set $h = f_{100} + d_1$. Applying (66) at the shared level-two vertices gives

$$f_{e_i} = h - d_i, \quad f_{e_i + e_j} = d_i + d_j - h.$$

It follows that q and $(d_1x_1 + d_2x_2 + d_3x_3 - h)^2$ have the same square coefficients and the same values at the six central vertices. Their difference is therefore

$$Ar(x), \quad r(x) = 1 - x_1 - x_2 - x_3 + x_1x_2 + x_1x_3 + x_2x_3.$$

For completeness, a multilinear quadratic vanishing at e_i has linear coefficients equal to the negative of its constant; vanishing at $e_i + e_j$ then makes every mixed coefficient equal to that constant. At a

central edge zero, the square has zero gradient and r has strictly positive inward derivatives. Thus nonnegativity forces $A \geq 0$. Since

$$r = (1 - x_1)(1 - x_2)(1 - x_3) + x_1x_2x_3,$$

this decomposes q into an affine square and an element of $\mathcal{D}_3$. Extremality would force q to be the excluded affine square.

For case (iii), choose the lower edge as 000–100 and the upper edge as 110–111. The three central edges are

$$001\text{--}101, \quad 001\text{--}011, \quad 010\text{--}011.$$

The face $x_2 = 0$ has opposite contact edges parallel to x_1 , so (67) gives $d_3 < d_1$. The face $x_2 = 1$ has opposite contact edges parallel to x_3 , giving $d_1 < d_3$, a contradiction.

For case (iv), take the lower edges as 000–100 and 000–010. The two central edges are 001–101 and 001–011, and the upper edge is 110–111. The two adjacent contacts on $x_3 = 0$ give

$$q(x_1, x_2, 0) = (h - d_1x_1 - d_2x_2)^2 + Tx_1x_2, \quad 0 < h < \min(d_1, d_2), \quad T \geq 0.$$

The corresponding contacts on $x_3 = 1$ give the same formula with h replaced by g , where $0 < g < \min(d_1, d_2)$. The mixed coefficient is unchanged, so the same T appears. Matching the square coefficient of x_3 now gives

$$q = (h + (g - h)x_3 - d_1x_1 - d_2x_2)^2 + Tx_1x_2 + \eta x_3(1 - x_3), \quad \eta = (g - h)^2 - d_3^2 \geq 0.$$

The last inequality follows from the two opposite x_1 contacts on $x_2 = 0$. On the proposed upper contact edge $x_1 = x_2 = 1$, the square is strictly positive, because the affine interpolation of h, g is less than $d_1 + d_2$. The other terms are nonnegative, a contradiction.

For case (v), choose the lower edges as 000–100 and 000–010, the upper edges as 011–111 and 110–111, and the central edge as 001–101. There are three contacts parallel to x_1 , at $(x_2, x_3) = (0, 0), (0, 1), (1, 1)$. Complete the square in x_1 :

$$q = d_1^2(x_1 - \alpha(x_2, x_3))^2 + R(x_2, x_3), \quad \alpha(x_2, x_3) = \alpha_0 - ux_2 - vx_3,$$

where $u, v > 0$ because the two mixed coefficients involving x_1 are positive. Each of the three recorded values of α lies in $(0, 1)$. Its fourth vertex value lies between $\alpha(1, 1)$ and $\alpha(0, 0)$, so α lies in $(0, 1)$ throughout the square. Consequently $R \geq 0$ there. On the lower contact edge $x_1 = x_3 = 0$, however, $\alpha(x_2, 0) > 0$, so the displayed square is strictly positive and q cannot vanish. This is again a contradiction.

E.6 Recovering the family from the remaining graph

Case (ii) is the only possibility left. For the representative with isolated edge 000–100, its central path is 110–010–011–001–101. Swap coordinates 1 and 3, then complement coordinates 1 and 2. This puts the contact edges in the following positions:

$$000\text{--}100, \quad 000\text{--}010, \quad 100\text{--}101, \quad 010\text{--}011, \quad 110\text{--}111.$$

The transformed quadratic need no longer have all mixed coefficients positive, but its square coefficients remain d_i^2 . Let $h = \sqrt{q(000)} > 0$. The first two contacts fix the edge roots $h/d_1, h/d_2 \in (0, 1)$, the constant h^2 , and the linear coefficients $-2hd_1, -2hd_2$. The next two contacts fix the roots

$(d_1 - h)/d_3$ and $(d_2 - h)/d_3$. Write the fifth root as $r/d_3 \in (0, 1)$, and set $D = d_1 + d_2 - h$ and $k = r - D$.

The derivative equations on the three vertical edges and the value on the edge through 110 determine the remaining coefficients:

$$a_3 = 2d_3(h + k), \quad c_{13} = -2d_3(d_1 + k), \quad c_{23} = -2d_3(d_2 + k), \quad c_{12} = 2d_1d_2 + k(2D + k).$$

Thus $q = q_{h,d,k}$. At $(h/d_1, 0, 0)$ the inward derivative in x_3 is $2d_3k(1 - h/d_1)$. This derivative is strictly positive by the first part of the proof, so $k > 0$. The fifth root gives $D + k < d_3$, while the first two give $0 < h < \min(d_1, d_2)$. These are exactly the strict five-contact assumptions.

Conversely, such a family member generates an exposed extreme ray by the five-contact argument. Its vertices have positive values: the contact edge endpoints are positive, and the remaining vertex 001 has value $(h + d_3)^2 + 2d_3k > 0$. Its mixed coefficient product is positive and each 2×2 principal determinant is negative, as follows from $D > 0$, $k > 0$, and the coefficient formulas above.

E.7 The affine-square rays

Finally let $q = \ell^2$ have positive values at every cube vertex. If the affine factor ℓ takes only one sign there, it is bounded away from zero on the cube. Then q is strictly positive and lies in the interior of $\mathcal{P}_3$, so its ray is not extreme. If ℓ takes both signs, its zero plane has a relatively open patch in the cube interior. In any decomposition $q = p + r$ with $p, r \in \mathcal{P}_3$, both summands vanish on this patch. Polynomial divisibility gives $p = \ell \ell_p$ for an affine ℓ_p . Nonnegativity on both sides of the plane forces ℓ_p to vanish on the patch; otherwise the product changes sign near a point where $\ell_p \neq 0$. Hence $\ell_p = \alpha \ell$, and evaluating away from the plane gives $\alpha \geq 0$. The same argument applies to r . Thus the ray of ℓ^2 is extreme. This completes the proof of Theorem 5.12.

F Stationary zeros of convex facet restrictions

This appendix supplies the zero-set argument used in Theorem 5.12. A stationary facet zero may lie in the facet interior or on one of its edges. The latter case is needed to exclude a vanishing inward derivative at an edge contact.

F.1 Subtracting an affine square

Lemma F.1. *Let $q \in \mathcal{P}_3$. If every zero of q lies in an affine plane $\ell(x) = 0$, with ℓ nonzero, then there is an $\epsilon > 0$ such that $q - \epsilon \ell^2 \in \mathcal{P}_3$. Consequently an extreme ray of $\mathcal{P}_3$ that is not an affine square has zeros affinely spanning $\mathbb{R}^3$.*

Proof. We use the rank-one subtraction criterion of Hildebrand [10, Lemma 4.3 in arXiv version 4]: for a copositive matrix B and a nonzero vector w , there exists an $\epsilon > 0$ with $B - \epsilon ww^\top$ copositive if and only if $w^\top u = 0$ for every nonnegative zero u of B .

Let $A \in \mathbb{S}^4$ be the homogeneous coefficient matrix of q , and let $T \in \mathbb{R}^{4 \times 8}$ have the columns $(1, v)$ for the cube vertices v . The cone generated by $(1, x)$ for $x \in C_3$ equals $T\mathbb{R}_+^8$. Thus $B = T^\top AT$ is copositive. Write $l \in \mathbb{R}^4$ for the homogeneous coefficient vector of ℓ and set $w = T^\top l$. The matrix T has rank four, so $w \neq 0$.

For a nonzero zero $u \geq 0$ of B , put $s = \mathbf{1}^\top u > 0$. There is an $x \in C_3$ with $Tu = s(1, x)$, and

$$0 = u^\top Bu = s^2 q(x).$$

Hence $w^\top u = l^\top Tu = s\ell(x) = 0$. The criterion gives $B - \epsilon ww^\top \in \text{COP}_8$. Every $(1, x)$ has a nonnegative vertex representation, so this says exactly that $q - \epsilon\ell^2 \geq 0$ on C_3 . For an extreme q , the decomposition $q = (q - \epsilon\ell^2) + \epsilon\ell^2$ forces q to be an affine square. $\square$

F.2 The stationary-facet lemma

Lemma F.2. *Let $q \in \mathcal{P}_3$ have positive square coefficients, positive mixed coefficients, and positive values at every cube vertex. Suppose a facet restriction has a positive-semidefinite quadratic coefficient matrix and a zero at which both facet derivatives vanish. If q generates an extreme ray, then it is an affine square.*

Proof. Simultaneously complementing all three coordinates preserves the mixed coefficients, so we may take the facet to be $z = 0$. Its stationary zero $u_0 = (x_0, y_0)$ is not a corner, because all vertex values are positive. It may be an edge point. We prove that all zeros of q lie in an affine plane, then apply Lemma F.1.

Positive-definite facet block. Write

$$q(u, z) = (u - u_0)^\top A(u - u_0) + 2zv^\top(u - u_0) + \gamma z + \tau z^2, \quad (68)$$

where

$$A = \begin{pmatrix} a & c \\ c & b \end{pmatrix} \succ 0, \quad a, b, c > 0, \quad v = (r, s)^\top > 0, \quad \tau > 0.$$

The inward derivative at the zero gives $\gamma \geq 0$. If the full quadratic coefficient matrix is PSD, (68) is a sum of affine squares and γz . Extremality and the positive square coefficients make it a single affine square. Otherwise its Schur complement

$$\kappa_0 = \tau - v^\top A^{-1}v$$

is negative.

For each $z \in [0, 1]$, let $u(z)$ be the unique minimizer of $q(u, z)$ over $[0, 1]^2$, and put $F(z) = q(u(z), z)$. Thus $F \geq 0$, $F(0) = 0$, and every zero of q has the form $(u(z), z)$ with $F(z) = 0$. We first describe this minimizer path.

When both coordinates are free, its velocity is

$$x'(z) = \frac{cs - br}{ab - c^2}, \quad y'(z) = \frac{cr - as}{ab - c^2}. \quad (69)$$

These velocities cannot both be positive. If both are nonpositive, one coordinate reaches zero and the other then decreases to zero. For example, on $x = 0$ the free coordinate has velocity $y' = -s/b$, and the gradient in the fixed coordinate increases with velocity $2(r - cs/b) \geq 0$. The lower bound therefore does not release.

Suppose instead that $x' > 0$. Then $cs > br$ and $y' < 0$. If y reaches zero first, x changes to velocity $-r/a$, and the lower-bound gradient in y increases with velocity $2(s - cr/a) > 0$. The path ends at 00. The strict inequality follows from

$$cs > br, \quad ab > c^2 \implies s > \frac{br}{c} > \frac{cr}{a}.$$

If x reaches one first, y continues with velocity $-s/b$. The upper-bound gradient in x decreases with velocity $2(r - cs/b) < 0$, so x remains one until y reaches zero. At 10, the lower-bound gradient in y increases with velocity $2s$, while the upper-bound gradient in x increases with velocity $2r$. The

latter may reach zero and release x . On the subsequent edge $y = 0$, x decreases with velocity $-r/a$ and y stays zero because $s - cr/a > 0$. Thus, up to relabeling, the possible paths are

$$\text{free} \longrightarrow x = 0 \longrightarrow 00, \quad \text{free} \longrightarrow x = 1 \longrightarrow 10 \longrightarrow y = 0 \longrightarrow 00. \quad (70)$$

Some phases may be absent.

An initial u_0 on a facet edge gives a truncation of these paths. An outward velocity at a lower bound immediately fixes that coordinate and makes the other decrease; the lower-bound gradient then increases. An outward velocity at an upper bound starts the upper-edge phase. An inward velocity starts the free phase. Simultaneous transitions omit a phase. If a free solution lies on a bound throughout a phase, either description gives the same formulas. The symmetric case $y' > 0$ follows by relabeling. This accounts for every initial position allowed by the hypotheses.

The path is continuous and piecewise affine. The value function is continuously differentiable and piecewise quadratic: on each phase the free gradients vanish, and the fixed coordinates have zero velocity, so $F'(z) = \partial_z q(u(z), z)$; the expressions agree at transitions. The curvature on each phase is given in Table 5.

Table 5: Curvature of the two-coordinate minimum in (68). The entries give $F''/2$.

Active coordinates	Curvature
Both free	$\kappa_0 = \tau - v^\top A^{-1}v < 0$
x fixed, y free	$\kappa_y = \tau - s^2/b$
y fixed, x free	$\kappa_x = \tau - r^2/a$
Both fixed	$\tau > 0$

For the upper-edge path in (70), $cs > br$ and $ab > c^2$ imply $r^2/a < s^2/b$, hence $\kappa_x > \kappa_y$. Its curvature sequence is therefore

$$\kappa_0, \quad \kappa_y, \quad \tau, \quad \kappa_x, \quad \tau, \quad \kappa_0 < 0, \quad \kappa_x > \kappa_y. \quad (71)$$

The lower-edge path has the shorter sequence κ_0, κ_y, τ , or its relabeling.

These sequences imply that the zeros of F for $z > 0$ are at most two isolated points or one interval. Here are the details. A negative-curvature piece cannot have an interior zero of a nonnegative function. Nor can a zero at an interior phase transition meet such a piece: differentiability gives $F' = 0$ at the zero, and the negative second derivative would give negative values immediately on that side. A zero at $z = 1$ can occur at the end of a concave piece, but is then the final zero.

If $\kappa_y \geq 0$, the tail starting with the first free-edge phase in (71) is convex. If $\kappa_y < 0 \leq \kappa_x$, the tail starting with the first corner is convex. In each case its nonnegative value function has a connected zero set. If $\kappa_x < 0$, both free-edge pieces are strictly concave. Only the two corner pieces can have interior zeros, and each has at most one because its curvature is positive. A terminal zero on a concave piece gives at most one later zero after the first corner, again for a total of two. The shorter path has the same simpler bound.

A zero interval requires one free-edge curvature to be zero. If $\kappa_y = 0$, then $\kappa_x > 0$, and the following positive-curvature corner leaves the interval with value and derivative zero and becomes strictly positive. Every later curvature is positive. If $\kappa_x = 0$, the preceding corner has its unique zero at the entry to the interval; the following corner is positive beyond its exit. Thus no separate zero accompanies a zero interval. The argument for a lower-edge path is the same. Across an interval, $u(z)$ is affine on one free-edge phase, so the corresponding zeros of q form a straight segment. The

only zero on $z = 0$ is $(u_0, 0)$. Its union with at most two points, or with one segment, is contained in an affine plane. This proves the lemma for a PD facet block.

Rank-one facet block. The positive diagonal and mixed coefficients give $A = (d_1, d_2)^\top (d_1, d_2)$ with $d_1, d_2 > 0$, and the stationary zero gives

$$q = (d_1x + d_2y - h)^2 + 2z(rx + sy + \eta) + \tau z^2, \quad r, s, \tau > 0. \quad (72)$$

Its base zero set is the segment $d_1x + d_2y = h$, $z = 0$. Vertex positivity gives $0 < h < d_1 + d_2$ and $h \neq d_1, d_2$. Put $\lambda_1 = r/d_1$, $\lambda_2 = s/d_2$, and first suppose $\lambda_1 < \lambda_2$.

For fixed $S = d_1x + d_2y$, minimizing $rx + sy$ uses coordinate x first. The minimum is

$$J(S) = \begin{cases} \lambda_1 S, & 0 \leq S \leq d_1, \\ \lambda_1 d_1 + \lambda_2 (S - d_1), & d_1 \leq S \leq d_1 + d_2. \end{cases}$$

For every $z > 0$ this allocation is unique. The remaining minimum is

$$F(z) = \min_{0 \leq S \leq d_1 + d_2} \{(S - h)^2 + 2z[J(S) + \eta] + \tau z^2\}. \quad (73)$$

If $h < d_1$, its path is $y = 0 \rightarrow 00$. If $h > d_1$, it is

$$x = 1 \rightarrow 10 \rightarrow y = 0 \rightarrow 00,$$

with weighted sums $h - \lambda_2 z$, d_1 , $h - \lambda_1 z$, and 0, respectively. Its curvature sequence is

$$\tau - \lambda_2^2, \quad \tau, \quad \tau - \lambda_1^2, \quad \tau, \quad \tau - \lambda_1^2 > \tau - \lambda_2^2. \quad (74)$$

The preceding argument gives at most two isolated outside zeros or one interval. Two isolated zeros could only be a first-corner zero followed by a later zero after a negative-curvature phase where x is free. If that phase had nonnegative curvature, the tail would instead be convex and its zero set connected.

This two-zero possibility is excluded by an inward derivative at the first corner. At a zero $(1, 0, z_A)$, vertex positivity gives $z_A \in (0, 1)$. The vertical edge restriction is a square, so, with $d_3 = \sqrt{\tau}$,

$$z_A = \frac{h - d_1}{d_3}.$$

The inward derivative in coordinate x is nonnegative, giving

$$d_1(d_1 - h) + rz_A \leq 0, \quad \text{and hence} \quad \lambda_1 \leq d_3.$$

Thus $\tau - \lambda_1^2 \geq 0$, a contradiction to the required negative curvature when x is free. The base segment is therefore accompanied by at most one outside point or one straight zero segment.

The base segment and a point lie in an affine plane. If an outside zero segment lies on $y = 0$, its restriction in (72) must be an affine square; coefficient comparison gives $r = d_1 d_3$ and $\eta = -h d_3$. Its zeros satisfy $d_1 x + d_3 z = h$. If it lies on $x = 1$, the corresponding comparison gives $s = d_2 d_3$ and $r + \eta = -(h - d_1) d_3$, and its zeros satisfy $d_2 y + d_3 z = h - d_1$. In either case the base segment and the outside segment lie in the same plane

$$d_1 x + d_2 y + d_3 z = h.$$

If $\lambda_2 < \lambda_1$, interchange the two coordinates.

It remains to consider $\lambda_1 = \lambda_2 = \lambda$. Now q depends on x, y only through S , and its minimizing sum is $S(z) = \max(h - \lambda z, 0)$. On the free-sum phase,

$$F(z) = 2(\eta + \lambda h)z + (\tau - \lambda^2)z^2.$$

An interior positive- z zero forces this polynomial to vanish identically: nonnegativity makes its derivative zero there, and its factor z leaves a linear factor. Then $\eta = -\lambda h$, $\tau = \lambda^2$, and q is the affine square $(S + \lambda z - h)^2$. Otherwise a free-sum zero can occur only at the final boundary $z = 1$, and all corresponding points lie in the plane $S + \lambda z = h$. After $S(z)$ reaches zero, the minimizer is 00; the resulting positive-curvature quadratic has at most one zero. A positive- z zero at an interior transition would likewise force the preceding free-sum polynomial to vanish identically, because F is differentiable there. A transition at $z = 1$ is a cube vertex and cannot be a zero. A free-sum zero at $z = 1$ leaves no later corner phase within the domain. Thus, in the non-square case, the base segment is accompanied by one point, or by one coplanar upper-facet segment. All zeros again lie in an affine plane.

Both possible ranks of the PSD facet block have now been covered. Lemma [F.1](#) completes the proof. $\square$